

# Interstratal differences in the tonology of Zacatepec Chatino: An argument for Harmonic Layer Theory

Eva Zimmermann, Leipzig University  
Eva.Zimmermann@uni-leipzig.de  
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## 1 Introduction

The tonal phonology of Zacatepec Chatino, an Eastern Otomanguean language, is highly interesting from a theoretical and typological perspective: It employs one of the typologically rarer cases of floating tones at the phrase layer<sup>1</sup> and the realization of phrasal floating tones often correlates with the appearance of mid tones whose source is mysterious at first glance. The data in (1) illustrates this phenomenon with two two-word combinations: In (1-a), the initial word has an underlying tonal melody consisting of two M-tones whereas the second word is tone-less underlyingly. On the surface, both moras of this second word are realized with an L-tone that can hence reasonably assumed to be a default tone. This same default L-tone can be found on the initial mora of the first word in (1-b): It has two moras and is specified for an M-tone and a floating H-tone that needs to dock to the following word. There are hence not enough underlying tones for the first mora and it is again realized as L-toned. The second word in (1-b) is underlyingly tone-less and realizes the floating H-tone of the first word on its final mora whereas the preceding mora is realized with an M-tone, marked in boldface. Since we saw in (1-b), that underlying M-tones don't spread across word boundaries, there is no underlying source for this M-tone and it again seems to be a default tone for an otherwise tone-less mora – but this seemingly contrasts with the default L-tone we saw in the other words.

(1) Floating H-tone realization at the phrase layer (Villard, 2015, 187)

| UNDERLYING |         |      | SURFACE REALIZATION |                |             |
|------------|---------|------|---------------------|----------------|-------------|
| a.         | /M.M/   |      | [M.M]               | [L.L]          |             |
|            | kwito   | kula | kwito               | kula           | 'old hen'   |
| b.         | /M (H)/ |      | [L.M]               | [ <b>M</b> .H] |             |
|            | kwana   | kula | nkwana              | kula           | 'old thief' |

In this paper, I argue that Zacatepec Chatino indeed employs two default tones, one being phonological and one being phonetic. Whereas the second word in (1-a) and the first word in (1-b) receive the phonological default tone by phonological L-epenthesis, the second word in (1-b) leaves the phonology with a tone-less initial mora that is then interpreted with a default M-tone in the phonetics. Such a pattern is predicted in a cyclic model where the phonological repair of L-epenthesis only applies at an early cycle of stem layer optimization but is unavailable at later layers. Moras that only become tone-less at the phrase layer to make room for floating tones (=word 2 in (1-b)) hence remain tone-less and are then interpreted with a phonetic default M-tone.

<sup>1</sup>I use the term 'layer' throughout and avoid the more common term 'stratum' or even 'level' since those are often associated with theoretical accounts based on co-grammars for each 'stratum' or 'layer' – a theoretical assumption that the proposal in this paper explicitly rejects.

This is a standard case of an interstratal asymmetry where different morphological layers show different phonological behaviour and is straightforwardly accounted for in stratal models that assume different phonologies for different morphological constituents (e.g. Kiparsky, 1982, 2015; Trommer, 2011, t.a.; Bermúdez-Otero, 2018, i.p.). However, such models face the conundrum that floating tone association at the phrase layer still needs to distinguish originally epenthetic L-tones from originally underlying L-tones since only the former are disassociated to make room for floating tones. In standard stratal models, however, the morphological affiliation of elements is assumed to be inaccessible at the next stratum ( $\sim$ Bracket Erasure (Chomsky and Halle, 1968; Kiparsky, 1982)).

I argue that this conundrum is resolved in a novel cyclic model of the morpho-phonology interface, namely Harmonic Layer Theory (Trommer, 2019; Zimmermann and Trommer, 2022; Zimmermann, 2023, 2025; Tebay, 2023; Tebay et al., 2025). It shares with standard stratal models the assumption that where morphologically more complex structures (=stems, words, phrase) are subsequently evaluated. In contrast to those models, however, it assumes that the same phonological grammar optimizes at each morphological layer. Interstratal asymmetries then arise since representations can predictably change across layers: They can incrementally decay in their activity which potentially results in different phonological behaviour. Returning to our introductory example in (1): Although L-epenthesis is in principle predicted for all otherwise tone-less moras, it ceases to be optimal at the phrase layer simply because the moras have decayed too much and their weak activity does not warrant epenthesis anymore. This model hence crucially relies on activity as an orthogonal property of phonological representations that can influence phonological behaviour (e.g. Smolensky and Goldrick, 2016; Faust and Smolensky, 2017; Faust, 2017; Zimmermann, 2018*a,b*, 2019, to appear; Rosen, 2018, 2019; Walker, 2020; Hsu, 2022).

The Harmonic Layer Theory account proposed in this paper also explains a variety of other interesting tonal facts about the language from purely phonological mechanisms: 1) The locality restriction of floating tones that need to dock to a word adjacent to their sponsoring morpheme reduces to a positional asymmetry between final and non-final moras, and 2) that tonal morphemes contrast floating tones that need to associate at the word and those that only associate at the phrase layer becomes a representational contrast between tones of different activities. The theoretical goal of this paper is hence two-fold: For one, it presents the first theoretical account of the complex Zacatepec Chatino tonal facts that relies on language-independent principles. And secondly, it argues for the novel Harmonic Layer Theory that predicts interstratal differences from a single phonological grammar for each language and is hence the more restricted model than standard stratal accounts. Finally, the account sheds new light on the discussion of the division of labor between phonetics and phonology. Two processes that are described as phonetic in Villard (2015) are argued to be better placed into the phonology since they predictably interact with other phonological processes. In addition, the co-existence of both a phonological and a phonetic default tone allows an explanation for the many mysterious mid tones whose distribution is captured by a set of language-specific realization rules in Villard (2015).

This paper is structured as follows: In section 2, the empirical facts about tones in Zacatepec Chatino are presented and an outlook of their theoretical analysis is given in 2.3. Section 3 presents the theoretical background assumption of Harmonic Layer Theory 3.1 and the adopted tonal representations 3.2 before the Harmonic Layer Theory analysis of the tonology of Zacatepec Chatino is presented in section 4. Section 5 concludes.

## 2 The Data: Tones in ZAC

Zacatepec Chatino (=ZAC) is a severely endangered Eastern Otomanguean language, spoken in the village of San Marcos Zacatepec in Oaxaca, Mexico. As most languages of the Otomanguean family, tone is pervasive in the morphology of the language (e.g. Palancar and Léonard, 2016). All the following data and generalizations are taken from Villard (2015) where a detailed description of the phonology and morphology of tone is provided, based on a meticulous phonetic study. I hence do not add any empirical findings; I only investigate the interesting pattern from a novel theoretical perspective. In subsection 2.1, some necessary background about tones and tonal morphemes in ZAC is given before the tonal processes at the phrase layer are discussed in subsection 2.2.

### 2.1 Tone melodies and tonal morphemes

The inventory of tones consists of the four level tones low (=L), mid (=M), high (=H), and superhigh (=S) and the two rising contours low-high (=LH) and low-superhigh (=LS). S-tones are exceptional since they have to be preceded by an L-tone on the first mora they are associated to and can only surface on their own on following moras. The tone-bearing unit (=TBU) in ZAC is the mora as can be concluded from the data in (2-d+e) where tonal sequences other than LH and LS surface on a long final vowel.<sup>2</sup> These are hence not true contour tones since each tone is associated to its own TBU. Note that long vowels are only possible in final syllables in ZAC. Tonal specifications are notated with superscripted tonal primitives in the following. Vowels without a superscript are not associated to an underlying tone at the word layer, notated as ‘X’ in the tonal specification. Their behaviour in the tonal phonology and their correct account is one of the main challenges this paper focusses on. Although Villard (2015) analyses them as tone-less, I present a different account in section 4 – for now, I use the neutral term ‘X-TBU’s’. The examples in (2-d) and (2-e) both end in a floating tone, given in brackets, their effect and source is also described in detail below.

#### (2) The mora as TBU in ZAC (Villard, 2015)

| Tones on V:# |                                  |      |                      |      |
|--------------|----------------------------------|------|----------------------|------|
| a.           | xa <sup>M</sup> a <sup>H</sup>   | M.H  | ‘luminosity’         | p.54 |
| b.           | pi <sup>L</sup> i <sup>LS</sup>  | L.LS | ‘fair skinned, pale’ | p.56 |
| c.           | xu <sup>M</sup> u <sup>L</sup> n | M.L  | ‘you will fight’     | p.65 |
| d.           | nkayoo <sup>M(H)</sup>           | X.M  | ‘s/he ground it’     | p.61 |
| e.           | rrsuu <sup>L(LS)</sup>           | X.L  | ‘message’            | p.68 |

The tone specifications of words are restricted to only 15 possible tonal melodies, given in (3) with their effect for bases with one, two, or three moras respectively. Cells given in grey are unattested words according to Villard (2015). As can be seen, the association of tonal melodies to TBU’s is governed by a preference for right-aligned tones and a dispreference for tone spreading. In a rule-based derivation, these tonal melodies would be taken to associate predictably in a 1:1 fashion, starting from the right edge. If a tonal melody contains more tones than TBU’s, tones at the left

<sup>2</sup>Villard (2015) gives most data in orthography and I follow this practice since it does not affect any of the tonal generalizations. An important convention is the use of a ‘coda’ /n/ which notates a nasal vowel. Underlyingly, nasality is only possible in final syllables and spreads across /w/ and /hw/ to the penultimate vowel.

edge thus remain unrealized.<sup>3</sup> For example, if the tonal melody M.H (cl.11) associates to a base with only a single mora, the initial M remains unrealized. Similarly, if there are more TBU's than tones, TBU's cluster at the right edge and the above-mentioned 'X-TBU's' emerge at the left edge. Even if we remain agnostic about their concrete nature for now, it is clear that no tone of the tonal morpheme associates to those TBU's.

Many imaginable combinations of tones are unattested as tonal melodies. For example, melodies contain maximally one H or S and these two can never co-occur in a melody. There is also dispreference for falling tonal melodies; something that will resurge later in the theoretical account. There are 5 tonal melodies that end in a floating tone. These are realized on a following word at the phrase layer, hence remain unassociated at the word layer. Villard (2015) lists the three floating tones H (cl.4), LS (cl.2+7) and L (cl.3+13) but discusses very different behaviours for them. Whereas floating H and LS associate predictably at the phrase layer, the putative floating L has a variety of seemingly unexpected surface effects on following words, very few of them actually involve realization of an additional L-tone. I hence propose that the floating L does not exist in ZAC and that the tonal melodies 3 and 13 are in fact L and M.LS.L. Since both tonal melodies result in a final associated L-tone at the word layer, the majority of unexpected tonal changes fall out as simple OCP effects between adjacent L-tones. Subsection 6.1.1 in the appendix discusses these tonal melodies in more detail and also shows how the theoretical account proposed in section 4 can be extended to those cases. Finally, note that the absence of any tone is also counted as a morpheme class (3-1).

(3) 15 tonal melodies in ZAC (Villard, 2015, 147+206)

| Class | Tonal morpheme | μ      | μ.μ      | μ.μ.μ      |
|-------|----------------|--------|----------|------------|
| 1.    | ∅              | X      | X.X      | X.X.X      |
| 2.    | L (LS)         | L (LS) | X.L (LS) | X.X.L (LS) |
| 3.    | L (L)          | L (L)  | X.L (L)  | X.X.L (L)  |
| 4.    | M (H)          | M (H)  | X.M (H)  | X.X.M (H)  |
| 5.    | M              | M      | X.M      | X.X.M      |
| 6.    | L.M            | M      | L.M      | X.L.M      |
| 7.    | L.M (LS)       | M (LS) | L.M (LS) | X.L.M (LS) |
| 8.    | LH             | LH     | X.LH     | X.X.LH     |
| 9.    | L.LS           | LS     | L.LS     | X.L.LS     |
| 10.   | M.M            | M      | M.M      | X.M.M      |
| 11.   | M.H            | H      | M.H      | X.M.H      |
| 12.   | M.M.L          | ML     | M.L      | M.M.L      |
| 13.   | M.LS.L (L)     | L (L)  | LS.L (L) | M.LS.L (L) |
| 14.   | M.H.M          | M      | H.M      | M.H.M      |
| 15.   | L.M.L          | L      | M.L      | L.M.L      |

<sup>3</sup>It is apparent that monomoraic class 12 words are a systematic exception to the generalization that only LH and LS are licit contours in Zacatepec Chatino. The inventory of contour tones might hence need reconsidering and an account probably needs to allow that ML – a possible but dispreferred contour – surfaces if an M remains otherwise unrealized. Since this formalization is orthogonal to the account proposed here, I leave this for future research.

The essential role that tone plays in the morphology of the language is best illustrated for verbs. An exemplifying paradigm for the verb *-u-lukwǎ* ‘to sweep’ is given in (4) which is realized with 5 different tonal melodies in different inflectional contexts. It can be seen that the 1SG, 2SG, and 3SG forms are segmentally identical in some aspects and are hence solely distinguished by tone. The plural forms are systematically identical to the 3SG form with the only difference that a person-specific plural enclitic follows. Note that the progressive prefix *nta<sup>M</sup>-* is special since it consistently bears a stable M-tone that is realized in addition to the tonal melody of the verb base – a composite tone sequence M.M.H hence results in the progressive forms in (4) (Villard, 2015, 270). These progressive forms are hence an exception to the claim that only the 15 tonal melodies in (2) are possible on words.

(4) Verbal inflection: verb */-u-lukwǎ/* ‘to sweep’ (Villard, 2015, 278)

|           | COMPL                                     | PROG                                                    | HAB                                       | POT                                      |
|-----------|-------------------------------------------|---------------------------------------------------------|-------------------------------------------|------------------------------------------|
| 1         | <i>nkalu<sup>L</sup>kwan<sup>LS</sup></i> | <i>nta<sup>M</sup>lu<sup>L</sup>kwan<sup>LS</sup></i>   | <i>ntulu<sup>L</sup>kwan<sup>LS</sup></i> | <i>kulu<sup>L</sup>kwan<sup>LS</sup></i> |
| 2         | <i>nkalu<sup>M</sup>kwa<sup>H</sup></i>   | <i>nta<sup>M</sup>lu<sup>M</sup>kwa<sup>H</sup></i>     | <i>ntulu<sup>L</sup>kwa<sup>M</sup></i>   | <i>kulu<sup>L</sup>kwa<sup>M</sup></i>   |
| 3         | <i>nkalukwa<sup>LH</sup></i>              | <i>nta<sup>M</sup>lu<sup>M</sup>kwa<sup>H</sup></i>     | <i>ntulukwa<sup>L(L)</sup></i>            | <i>kulukwa<sup>L(L)</sup></i>            |
| 1.PL.INCL | <i>nkalukwa<sup>LH</sup> nan</i>          | <i>nta<sup>M</sup>lu<sup>M</sup>kwa<sup>H</sup> nan</i> | <i>ntulukwa<sup>L(L)</sup> nan</i>        | <i>kulukwa<sup>L(L)</sup> nan</i>        |
| 1.PL.EXCL | <i>nkalukwa<sup>LH</sup> wa</i>           | <i>nta<sup>M</sup>lu<sup>M</sup>kwa<sup>H</sup> wa</i>  | <i>ntulukwa<sup>L(L)</sup> wa</i>         | <i>kulukwa<sup>L(L)</sup> wa</i>         |
| 2.PL      | <i>nkalukwa<sup>LH</sup> wan</i>          | <i>nta<sup>M</sup>lu<sup>M</sup>kwa<sup>H</sup> wan</i> | <i>ntulukwa<sup>L(L)</sup> wan</i>        | <i>kulukwa<sup>L(L)</sup> wan</i>        |
| 3.PL      | <i>nkalukwa<sup>LH</sup> ne?</i>          | <i>nta<sup>M</sup>lu<sup>M</sup>kwa<sup>H</sup> ne?</i> | <i>ntulukwa<sup>L(L)</sup> ne?</i>        | <i>kulukwa<sup>L(L)</sup> ne?</i>        |

However, this is only one of many possible tonal paradigms for verbal inflection. For illustration, two more abstract paradigms are given in (5) which only give the tonal melodies for each context. Since the tonal specification of the plural never contrasts, only the singular forms are shown. The potential and habitual are also conflated since their tones are always identical. The choice for one or another tonal inflection pattern is phonologically and semantically unpredictable, I’ll refer to these as conjugational classes in the following.

(5) Verbal inflection: Some more conjugation classes

| ‘pluck it’ (pp.367,400,410) |         |            |         | ‘peel it’ (pp.385,392,411) |          |            |          |
|-----------------------------|---------|------------|---------|----------------------------|----------|------------|----------|
|                             | COMPL   | PROG       | HAB/POT |                            | COMPL    | PROG       | HAB/POT  |
| 1SG                         | X (1)   | M.M.L (12) | X (1)   | 1SG                        | X (1)    | M (5)      | X (1)    |
| 2SG                         | L.M (6) |            |         | 2SG                        | M.M (10) | M.H.M (14) | M.M (10) |
| 3SG                         | X (1)   | M.M.L (12) | X (1)   | 3SG                        | M.H (11) |            | L.LS (9) |

The tonal specification of all surface verb forms is hence co-determined by their arbitrary lexical conjugation feature and inflectional features. I hence assume that the tonal melody of verbs is only determined at the word layer where inflectional features are present. For each verb, a tonal morpheme expressing both inflectional features and specific conjugation features is then added to the so-far tone-less verb root.<sup>4</sup> A parallel assumption is made for inalienable nouns and predicative ad-

<sup>4</sup>As Villard (2015) meticulously works out in chapter 5, the tonal pattern of the 3SG.COMPL allows to predict most of the tonal melodies of other inflectional forms. An alternative conclusion could hence be that the 3SG.COMPL shows the underlying tonal specification of verbal roots and the other tonal morphemes are phonologically predictable suppletive allomorphs that are chosen based on the underlying tonal melody. In terms of lexical storage and the transfer of lexical information into the morphology, this suppletive allomorphy account and the one based on conjugational

jectives which are also obligatorily inflected for person and this inflection is expressed tonally. The theoretical account of this paper in 4 derives these morphological contexts which receive their tonal melody at the word layer. In section 6.1.2 in the appendix, however, it is shown that the analysis is also perfectly compatible with the assumption that some uninflected word classes (e.g. alienable nouns, adverbs, non-predicative verbs, and some particles) already receive their tonal specification at the stem layer.

## 2.2 Tonal processes at the phrase layer

After having established the presence of multiple tonal morphemes in ZAC, we now return to the phonological behaviour of tones and take a closer look at the ‘X’-TBU’s we saw in (3), namely those that did not receive a tone from a tonal melody at the word layer. Their fate depends on potentially preceding words and can hence only be determined at the phrase layer. In the default case, X-TBU’s are realized with an L-tone. This is shown in (6) where the surface tone realizations for some two-word combinations are given. The second line of each datum repeats the tonal melody specification for each word and the phrasal surface tones to ease readability. It can be seen that all X-TBU’s are realized with a surface-L in (6).

(6) Default L-tones (Villard, 2015, 187)

|    | WORD TONES                                                         | PHRASAL SURFACE                                                                              |                       |
|----|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------|
| a. | kwana kula<br>/X.X/ /X.X/                                          | kwa <sup>L</sup> na <sup>L</sup> ku <sup>L</sup> la <sup>L</sup><br>[L.L][L.L]               | ‘old mirror’          |
| b. | nkanan <sup>M</sup> kwila<br>/X.M/ /X.X/                           | nka <sup>L</sup> nan <sup>M</sup> kwi <sup>L</sup> la <sup>L</sup><br>[L.M][L.L]             | ‘I looked for fish’   |
| c. | ka <sup>L</sup> kwen <sup>M</sup> kwila<br>/L.M/ /X.X/             | ka <sup>L</sup> kwen <sup>M</sup> kwi <sup>L</sup> la <sup>L</sup><br>[L.M][L.L]             | ‘you will vomit fish’ |
| d. | kwi <sup>M</sup> to <sup>M</sup> kula<br>/M.M/ /X.X/               | kwi <sup>M</sup> to <sup>M</sup> ku <sup>L</sup> la <sup>L</sup><br>[M.M][L.L]               | ‘old hen’             |
| e. | jn <sup>y</sup> ã <sup>M</sup> ke <sup>L</sup> kula<br>/M.L/ /X.X/ | jn <sup>y</sup> ã <sup>M</sup> ke <sup>L</sup> ku <sup>L</sup> la <sup>L</sup><br>[M.L][L.L] | ‘old hen of yours’    |

Any X-TBU following a word-final H or S will also be realized with H (7-a+b) and S (7-c-e) respectively. This is taken to be due to tonal spreading that is only possible for these two tones. That word-final M-tones cannot spread was already apparent in (6-b-d) where X-TBU’s following an M are realized with L. For a word-final L, it is impossible to tell L-spreading apart from default L-insertion since both result in L-toned TBU’s. The account in section 4 follows Villard (2015) and restricts spreading to H- and S-tones. Example (7-f) adds one example which shows that spreading never applies if a toned TBU follows.

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classes I assume are more or less equivalent.

(7) Spreading of H and S (Villard, 2015, 184+187)

|    | WORD TONES                                                                                            | PHRASAL SURFACE                                                                                                   |                  |
|----|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------|
| a. | kwi <sup>M</sup> na <sup>H</sup> kula<br>/M.H/ /X.X/                                                  | kwi <sup>M</sup> na <sup>H</sup> ku <sup>H</sup> la <sup>H</sup><br>[M.H][H.H]                                    | ‘old snake’      |
| b. | yanan <sup>LH</sup> kula<br>/X.LH/ /X.X/                                                              | ya <sup>L</sup> nan <sup>LH</sup> ku <sup>H</sup> la <sup>H</sup><br>[L.LH][H.H]                                  | ‘old corn cob’   |
| c. | yu <sup>L</sup> sin <sup>LS</sup> kula<br>/L.LS/ /X.X/                                                | yu <sup>L</sup> sin <sup>LS</sup> ku <sup>S</sup> la <sup>S</sup><br>[L.LS][S.S]                                  | ‘old sea turtle’ |
| d. | tit <sup>y</sup> uk <sup>L</sup> wa <sup>LS</sup> naten <sup>L</sup><br>/X.L.LS/ /X.L/                | ti <sup>L</sup> t <sup>y</sup> uk <sup>L</sup> wa <sup>LS</sup> na <sup>S</sup> ten <sup>L</sup><br>[L.L.LS][S.L] | ‘twelve people’  |
| e. | tit <sup>y</sup> uk <sup>L</sup> wa <sup>LS</sup> kwana <sup>M</sup><br>/X.L.LS/ /X.M/                | ti <sup>L</sup> t <sup>y</sup> uk <sup>L</sup> wa <sup>LS</sup> kwa <sup>S</sup> na <sup>M</sup><br>[L.L.LS][S.M] | ‘twelve thieves’ |
| f. | tit <sup>y</sup> uk <sup>L</sup> wa <sup>LS</sup> mu <sup>L</sup> l <sup>y</sup> LS<br>/X.L.LS/ /L.M/ | ti <sup>L</sup> t <sup>y</sup> uk <sup>L</sup> wa <sup>LS</sup> kwa <sup>L</sup> na <sup>M</sup><br>[L.L.LS][L.M] | ‘twelve mules’   |

This spreading is unbounded and can affect any number of adjacent X-TBU’s, even across words. In (8-a), a word ending in an LS tone is followed by multiple words consisting only of X-TBU’s and the S spreads to all of these on the surface. Example (8-b) contrasts the same sentence with a different initial word ending in M: As we expect, all following X-TBU’s are realized with L since M cannot spread.

(8) S-spreading across words (Villard, 2015, 159)

- a. /yu<sup>L</sup>sin<sup>LS</sup> nkayako tsaka lijya kula nd<sup>y</sup>ika: tsa:n/  
[yu<sup>L</sup>sin<sup>LS</sup> nka<sup>S</sup>ya<sup>S</sup>ko<sup>S</sup> tsa<sup>S</sup>ka<sup>S</sup> li<sup>S</sup>jya<sup>S</sup> ku<sup>S</sup>la<sup>S</sup> nd<sup>y</sup>i<sup>S</sup>ka: tsa:n<sup>S</sup>]  
‘The turtle ate an old sugar cane every day.’  
/L.LS/ /X.X.X/ /X.X/ /X.X/ /X.X/ /X.X.X/ /X.X/  
[L.LS] [S.S.S] [S.S] [S.S] [S.S] [S.S.S] [S.S]
- b. /kãn<sup>?</sup> nkayako tsaka lijya kula nd<sup>y</sup>ika: tsa:n/  
[kãn<sup>?</sup> nka<sup>L</sup>ya<sup>L</sup>ko<sup>L</sup> tsa<sup>L</sup>ka<sup>L</sup> li<sup>L</sup>jya<sup>L</sup> ku<sup>L</sup>la<sup>L</sup> nd<sup>y</sup>i<sup>L</sup>ka: tsa:n<sup>L</sup>]  
‘That one ate an old sugar cane every day.’  
/M/ /X.X.X/ /X.X/ /X.X/ /X.X/ /X.X.X/ /X.X/  
[M] [L.L.L] [L.L] [L.L] [L.L] [L.L.L] [L.L]

The third way in which X-TBU’s can receive a tone are floating tones. Recall from above that I disregard the putative floating L for now and focus only on the floating H and LS observed in tonal classes 1, 4, and 7. They remain unassociated during the tone melody association at the word layer and only find their host at the phrase layer. This host has to be the rightmost X-TBU of a following word, shown for floating H-tones in (9) and floating LS-tones in (10) where a word sponsoring such a floating tone is followed by one other word. Examples a. start with a context where the floating tones are followed by a word that is tonally specified and hence has no X-TBU. The floating tones

remain without any surface effect in this case. In examples b., a word with a single X-TBU follows and the floating tones are realized on this TBU that becomes H- (9-b) and S-toned (10-b) respectively. Examples c-e are particularly interesting since they involve multiple X-TBU's that follow the floating tone. As can be seen, the H and LS is only realized on the rightmost of these TBU's. The floating tones hence do not associate to all following TBU's, something one might expect given their ability to spread (cf. (7) and (8)). In contrast, all TBU's preceding the landing site of a floating tone are realized with an M-tone. This is surprising given the generalizations so far: If spreading of H and S is impossible, we would expect a default-L. At this point I want to emphasize again that Villard (2015) provides a careful and detailed phonetic study of the tonal processes and proves that this tone is indeed systematically different from both L, H, or S (cf. especially pp.165-168).

(9) Floating H at the phrase layer (Villard, 2015, 187+233)

|    | WORD TONES                                                                              | PHRASAL SURFACE                                                                   |                              |
|----|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------|
| a. | kwa <sup>M(H)</sup> nta <sup>M</sup> sa <sup>H</sup> la <sup>M</sup><br>/M (H)/ /M.H.M/ | kwa <sup>M</sup> nta <sup>M</sup> sa <sup>H</sup> la <sup>M</sup><br>[M][M.H.M]   | 'already you are opening it' |
| b. | kwa <sup>M(H)</sup> nkasa <sup>L</sup> lo <sup>M</sup><br>/M (H)/ /X.L.M/               | kwa <sup>M</sup> nka <sup>H</sup> sa <sup>L</sup> lo <sup>M</sup><br>[M][H.L.M]   | 'already you threw it away'  |
| c. | kwana <sup>M(H)</sup> kula<br>/X.M (H)/ /X.X/                                           | nkwa <sup>L</sup> na <sup>M</sup> ku <sup>M</sup> la <sup>H</sup><br>[L.M][M.H]   | 'old thief'                  |
| d. | kwa <sup>M(H)</sup> nkajilyan <sup>M</sup><br>/M (H)/ /X.X.M/                           | kwa <sup>M</sup> nka <sup>M</sup> ji <sup>H</sup> lyan <sup>M</sup><br>[M][M.H.M] | 'already I farted'           |
| e. | kwa <sup>M(H)</sup> nkajinyan<br>/M (H)/ /X.X.X/                                        | kwa <sup>M</sup> nka <sup>M</sup> ji <sup>M</sup> nyan <sup>H</sup><br>[M][M.M.H] | 'already I asked for it'     |

(10) Floating LS at the phrase layer (Villard, 2015, 187+233)

|    | WORD TONES                                                                                                | PHRASAL SURFACE                                                                                      |                            |
|----|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------|
| a. | ti <sup>y</sup> a <sup>L(LS)</sup> nta <sup>M</sup> sa <sup>H</sup> la <sup>M</sup><br>/X.L (LS)/ /M.H.M/ | ti <sup>L</sup> ya <sup>L</sup> nta <sup>M</sup> sa <sup>H</sup> la <sup>M</sup><br>[L.L][M.H.M]     | 'early you are opening it' |
| b. | ti <sup>y</sup> a <sup>L(LS)</sup> nkasa <sup>L</sup> lo <sup>M</sup><br>/X.L (LS)/ /X.L.M/               | ti <sup>L</sup> ya <sup>L</sup> nka <sup>LS</sup> sa <sup>L</sup> lo <sup>M</sup><br>[L.L][LS.L.M]   | 'early you threw it away'  |
| c. | mul <sup>L</sup> ya <sup>M(LS)</sup> kula<br>/M.L (LS)/ /X.X/                                             | mul <sup>L</sup> ya <sup>M</sup> ku <sup>M</sup> la <sup>LS</sup><br>[L.M][M.LS]                     | 'old mule'                 |
| d. | ti <sup>y</sup> a <sup>L(LS)</sup> nkajilyan <sup>M</sup><br>/X.L (LS)/ /X.X.M/                           | ti <sup>L</sup> ya <sup>L</sup> nka <sup>M</sup> ji <sup>LS</sup> lyan <sup>M</sup><br>[L.L][M.LS.M] | 'early I farted'           |
| e. | ti <sup>y</sup> a <sup>L(LS)</sup> nkajinyan<br>/X.L (LS)/ /X.X.X/                                        | ti <sup>L</sup> ya <sup>L</sup> nka <sup>M</sup> ji <sup>M</sup> nyan <sup>LS</sup><br>[L.L][M.M.LS] | 'early I asked for it'     |

The data in (11) illustrates an important locality condition: Floating tones must associate to the immediately following word, even if they are more X-TBU's following this one. Floating tone association is hence not as non-local as tone spreading that happily crosses multiple word boundaries (cf. (8)). Interestingly enough, originally floating tones can spread to multiple following words once they are associated. In derivational terms, floating tone association can hence feed H-/S-spreading.



This is shown in (11) where a floating H and LS precede multiple words, all consist exclusively of X-TBU's. The tones dock to the rightmost X-TBU of the following word and spread to all following ones. The two X-TBU's preceding its landing site are once again realized with an M-tone.

(11) Floating tone association and H/S-spreading (Villard, 2015, 166+168)

- a. /kin<sup>y</sup>i<sup>M</sup>(<sup>H</sup>) nkayako tsaka lijya kula nd<sup>y</sup>ika: tsa:n/  
 [ki<sup>L</sup>n<sup>y</sup>i<sup>M</sup> nka<sup>M</sup>ya<sup>M</sup>ko<sup>H</sup> tsa<sup>H</sup>ka<sup>H</sup> lij<sup>H</sup>ya<sup>H</sup> ku<sup>H</sup>la<sup>H</sup> nd<sup>y</sup>i<sup>H</sup>ka: tsa:<sup>H</sup>]  
 'The bird ate an old sugar cane every day.'  
 /X.M (H)/ /X.X.X/ /X.X/ /X.X/ /X.X/ /X.X.X/ /X.X/  
 [L.M] [M.M.H] [H.H] [H.H] [H.H] [H.H.H] [H.H]
- b. /naten<sup>L</sup>(<sup>LS</sup>) nkayako tsaka lijya kula nd<sup>y</sup>ika: tsa:n/  
 [na<sup>L</sup>ten<sup>L</sup> nka<sup>M</sup>ya<sup>M</sup>ko<sup>LS</sup> tsa<sup>S</sup>ka<sup>S</sup> lij<sup>S</sup>ya<sup>S</sup> ku<sup>S</sup>la<sup>S</sup> nd<sup>y</sup>i<sup>S</sup>ka: tsa:n<sup>S</sup>]  
 'People ate an old sugar cane every day.'  
 /X.L (LS)/ /X.X.X/ /X.X/ /X.X/ /X.X/ /X.X.X/ /X.X/  
 [L.L] [M.M.LS] [S.S] [S.S] [S.S] [S.S.S] [S.S]

### 2.3 In a nutshell: A summary of the data and an outlook of the theoretical proposal

The tonal processes discussed above can best be summarized if one recalls the different outcomes for X-TBU's in different contexts, listed in (12). For one, they can be the landing site for a floating H or LS tone hosted by a preceding word if they are the rightmost X-TBU in the word following this floating tone (12-a). Similarly, if they follow an associated H- or S-tone, they are realized as H or S respectively if no toned TBU intervenes (12-b). Crucially, this spreading applies across any number of X-TBU's and word boundaries. If an X-TBU precedes the landing site of a floating tone, however, it is realized with an M-tone (12-c). If none of the above mentioned contexts are given, an X-TBU is realized with an L-tone (12-d). Interestingly, an X-TBU can be realized with nearly all possible tonal specifications of the language, namely L, M, H, S, or LS. Only the contour LH is excluded on an X-TBU.

(12) Summary: The fate of X-TBU's at the phrase layer

|                              |                    |       |
|------------------------------|--------------------|-------|
| a. Floating tone realization | /M (H)/ /X/        | [H]   |
|                              | /M (LS)/ /X/       | [LS]  |
| b. H/S-Spreading             | /H/ /X/            | [H]   |
|                              | /LS/ /X/           | [S]   |
|                              | /... (H)/ /X/ /X/  | [H]   |
|                              | /... (LS)/ /X/ /X/ | [S]   |
| c. M-realization             | /M (H)/ /X.X/      | [M.H] |
| d. Default L                 | /M/ /X/            | [L]   |
|                              | /L/ /X/            | [L]   |
|                              | /X/ /X/            | [L]   |

The main theoretical challenge arising from this pattern is the appearance of M-tones on X-TBU's that precede the landing site of a floating tone (13-a). As was already emphasized, we would expect that these X-TBU's are realized with a default L if no spread of a preceding H or S is possible. A second property of the system that requires a theoretical explanation is the ambivalent behaviour of floating tones when it comes to locality: They can skip multiple potential hosts and only associate to the rightmost following TBU. However, this TBU has to be the rightmost TBU of the following word; a floating tone can hence seemingly only cross one word boundary but not two (13-b). And a third theoretical challenge of the ZAC tonal morpho-phonology can be found in the inventory of tonal melodies (3) itself. The way tonal melodies are realized on their base is in principle completely predictable since they always associate 1:1 from right to left. If we recall that verbal inflectional is expressed by different tonal melodies, one can hence conclude that floating tonal melodies as exponents of inflectional features are added to verbs at the word layer and associate predictably. There is a problem with such a seemingly simply account, namely the fact that tonal melodies can contrast floating and non-floating final tones that are only realized at the phrase layer on the following word (13-c). This is most apparent for melodies 4 /M (H)/ and 11 /M.H/: They both contain the same tones but in one case, the final tone waits with its association until a following word is added whereas the other associates straight away to the final TBU of its base.

(13) The challenging empirical generalizations

- a. Every X-TBU that precedes a TBU associated to an underlyingly floating tone is realized with M
- b. Floating tones associate to the rightmost X-TBU of the following word
- c. Tonal melodies associate predictably to their base but distinguish floating and non-floating tones

In the following, I give a brief overview of how these generalizations are accounted for in the cyclic model I propose in section 4. Firstly, I argue that the mysterious M-tone (13-a) is the phonetic default realization of a phonologically toneless TBU. Those TBU's result from an asymmetry: L-epenthesis applies at the stem layer but not at the phrase layer. TBU's that only become tone-less at this late stage of the derivation (=to make room for floating tones) hence remain tone-less. In Harmonic Layer Theory, this interstratal difference results from a predictable weakening of TBU's across layers, not the assumption of different phonological grammars at different morpho-syntactic domains. At the phrase layer, TBU's are simply so weak, that their need to be associated to a tone is not important enough to warrant L-epenthesis, even though the same phonological grammar optimizes both the stem and the phrase layer.

That floating H-and S-tones associate to the rightmost X-TBU of a following word (12) follows from a combination of three phonological factors: Firstly, floating H/S can only overwrite originally epenthetic tones, never originally underlying tones. In the HLT account, this follows since tone epenthesis is always minimal and results in a very weak tone that is easily deleted; in contrast to stronger underlying tones. Secondly, sequences of H/S and L-tones are dispreferred in ZAC, the realization of floating H/S will hence cause deletion of all following weak L-tones. And thirdly, word-final TBU's are taken to be strong positions that are exempt from the predictable activity decay in ZAC – word-final TBU's can hence not remain tone-less but must serve as the docking site of a floating H/S tone.

The last challenge of contrasting floating and non-floating tones is explained as a representational contrast for tonal activity. Some tonal melodies happen to end in a tone with an exceptionally high activity. Since there is a ban against too much activity, these tones cannot associate at the word layer where the rest of the tonal melody finds its host. They hence have to ‘wait’ until the phrase layer where the predictable decay for all tones has finally weakened them enough to be able to associate.

### 3 The theory: Background assumptions

#### 3.1 Harmonic Layer Theory

Harmonic Layer Theory (=HLT; Trommer, 2019; Zimmermann and Trommer, 2022; Zimmermann, 2023, 2025; Tebay, 2023; Tebay et al., 2025) is a novel theory of the phonology-morphology interface that rests on three background assumptions, listed in (14).

(14) Harmonic Layer Theory

a. **Cyclicity and morphological layers**

The phonology optimizes three sub-sequent layers of increasing morphological complexity: The stem, word, and phrase layer.

b. **One grammar**

A single phonological grammar evaluates each layer.

c. **Gradient Symbolic Representations (=GSR)**

All phonological elements have a certain activity that can gradiently differ and result in gradient constraint violations.

The first assumption is of course reminiscent of the basic architecture in Lexical Phonology (Kiparsky, 1982, 1985; Kaisse and Shaw, 1985; Mohanan, 1986) and its OT-descendant Stratal OT (Kiparsky, 2015; Trommer, 2011, t.a.; Bermúdez-Otero, 2018, i.p.). The second assumption crucially sets HLT apart from these theories. In both Lexical Phonology and Stratal OT, potentially different phonological grammars are active at the different morpho-syntactic layers. In contrast, HLT relies on a single phonological grammar that cyclically optimizes stems, words, and phrases. Interstratal differences where different phonological behaviour is active at different layers solely results from the third assumption (14-c), namely the fact that all phonological elements have a certain activity (Smolensky and Goldrick, 2016). Originally, GSR were proposed to account for lexical differences: If certain morphemes show an unexpected phonological behaviour, this falls out since they happen to have a phonological element with an exceptional activity in their underlying representation (Zimmermann, 2018*b,a*, 2019; Rosen, 2018, 2019; Kushnir, 2019; Walker, 2020). This is formally modeled via gradient constraint violations. Gradient markedness and all faithfulness constraints are hence violated relativized to the activity of the respective element. An underlyingly weaker element, for example, will thus be easier to delete than a stronger element. HLT utilizes another prediction of gradient representations, namely the fact that activity can be adjusted in the phonology. The activity adjustment relevant in the following is decay: Elements predictably lose activity to decrease markedness. How activity adjustment as a phonological process arises from gradient constraints is illustrated in (15) for tones. A context-free markedness constraint \*T (15-a) penalizes any tonal

activity.<sup>5</sup> Realization of any fully active input tone as in candidate (16-a) thus induces -1 violation of this constraint. Deletion of a tone as in (16-b) consequently avoids any violation of \*T for this tone. In the optimization in (16), however, this strategy is excluded by the inviolable weight of MAX-T (15-b). Candidates (16-c) and (16-d) reduce the violations of \*T by decreasing the input activity rather than completely deleting the tones. Any change in the level of input activity only induces a violation of the faithfulness constraint IDENTACT-T (15-c). Near-complete reduction to a maximally weak tone  $T_{0.1}$  in (16-d) is preferred when it comes to \*T but is excluded in this optimization by the inviolable weight of the threshold constraint IDENTACT-T[ $\geq 0.1$ ] (15-d) that penalizes any activity change that exceeds 0.1. Decay of every input tone by 0.1 activity in (16-d) hence emerges as optimal. As all models based on gradient representation, the phonological optimization in the following is based on Harmonic Grammar and thus weighted constraints (Legendre et al., 1991, 2006; Pater, 2009; Potts et al., 2010). In such models, a harmony score ( $\mathcal{H}$ ) is calculated for each candidate that adds all constraint violations times their respective constraint weights and the candidate with the harmony score closest to zero is optimal. The constraint weights in (16) already illustrate a practice I adopt that hopefully helps to ease readability: All constraints that are never violated in a winning candidate get the drastic weight of ‘ $\infty$ ’.

- (15) a. \*T W=8  
Assign -x violation for every tone tonal root node  $\bullet_x$ .
- b. MAX-T W= $\infty$   
Assign -x violation for every input tonal root node  $\bullet_x$  without an output correspondent.
- c. IDENTACT-T W=1  
For every tonal root node  $\bullet_y$  corresponding to input  $\bullet_x$ , assign  $-(|x-y|)$  violation.
- d. IDENTACT-T[ $\geq 0.1$ ] W= $\infty$   
For every tonal root node  $\bullet_y$  corresponding to input  $\bullet_x$ , assign  $-(|x-y|)$  violation if  $(|x-y|) > 0.1$ .
- (16) Activity loss for tones

| $M_1 H_1$              | IDENTACT-T[ $\geq 0.1$ ]<br>$\infty$ | MAX-T<br>$\infty$ | *T<br>8 | IDENTACT-T<br>1 | $\mathcal{H}$ |
|------------------------|--------------------------------------|-------------------|---------|-----------------|---------------|
| a. $M_1 H_1$           |                                      |                   | -2      |                 | -16.0         |
| b. $M_1$               |                                      | -1                | -1      |                 | $-\infty$     |
| c. $M_{0.1} H_{0.1}$   | -1.8                                 |                   | -0.2    |                 | $-\infty$     |
| ☞ d. $H_{0.9} M_{0.9}$ |                                      |                   |         | -0.2            | -14.4         |

In a system based on gradient representations, activity adjustment is hence an expected phonological process. And in a cyclic model where a grammar like (16) optimizes increasingly larger morphological domains, elements are predicted to gradually adjust their activity across layers, as shown in (17).

<sup>5</sup>All constraints in the following are defined over ‘tonal root nodes’ instead of ‘tones’ – an assumption consistent with the remaining account of ZAC that is introduced in more detail in subsection 3.2.

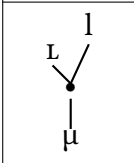
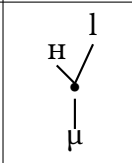
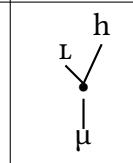
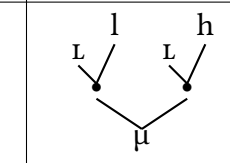
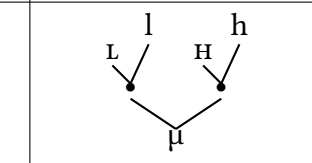
- (17) Activity decay across layers, predicted from grammar (16)
- |                       |                                                 |                                                 |
|-----------------------|-------------------------------------------------|-------------------------------------------------|
| Optimization stem ①   | $/M_1 H_1/ \rightarrow M_{0.9} H_{0.9}$         |                                                 |
| Optimization word ②   | $/M_{0.9} H_{0.9}/ \rightarrow M_{0.8} H_{0.8}$ |                                                 |
| Optimization phrase ③ |                                                 | $/M_{0.8} H_{0.8}/ \rightarrow M_{0.7} H_{0.7}$ |

Seemingly identical phonological contexts might thus result in different phonological behaviour at different morphological layers since the contexts are created by elements with different activity. The same grammar hence predicts interstratal differences in HLT from predictable activity adjustment: a mechanism that will become more apparent in the case study of ZAC in section 4.

### 3.2 Tonal representations

In ZAC, H- and S-tones behave systematically different from M- and L-tones: Only H and S spread at the phrase layer and only H and S persist as floating tones across layers and thus patiently wait until they can associate. M and L, on the other hand, are immediately deleted if they don't find a TBU at the very layer that they are added to the structure. To capture this natural class distinction, I assume that tones are not primitive elements but have a complex internal featural structure. The assumption that tones are composed of hierarchically organized subtonal features has been around in various specific incarnations since the early autosegmental works (e.g. Yip, 1980; Clements, 1981; Hyman, 1985; Pulleyblank, 1986; Inkelas, 1989). Relatively recent new empirical evidence for such models can be found in McPherson (2017) or Meyase (2021). The specific subtonal model I assume is Register Tier Theory where tones are composed of a high or low register feature and a high or low melody feature, resulting in maximally four contrastive tone levels (Snider, 1998, 2020). These features are organized on different tiers and associate to a tonal root node. The latter is given as • in the following and the former are abbreviated with lowercase 'h - l' for register and capital 'H - L' for melody features respectively. The resulting 5 tone specifications possible for TBU's in ZAC are given in (18). Both H- and S-tones are root nodes associated to a high register features 'h' and only differ in their melody feature. L- and M-tones, on the other hand, are root nodes associated to a low register feature 'l'. The only contour tones possible in ZAC are the two rising contours LH and LS and those are taken to consist of two different tonal root nodes associated to a single TBU. The activity differences that are relevant for the account in section 4 all pertain to the tonal root node. That activity differences for tonal features might allow an extension to more complex patterns in ZAC is discussed in section 6.1.1 in the appendix.

- (18) Register Tier Theory tone representations for ZAC

| Low (L)                                                                             | Mid (M)                                                                             | High (H)                                                                            | Low-High (LH)                                                                       | Low-Superhigh (LS)                                                                   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |  |  |  |

In the account below, the constraints against tone spreading and the ones against deletion and de-association of tones are relativized to high/low register features. Nothing crucial in the following, however, hinges on that and the account can easily be re-implemented in a model where all con-

straints refer to tonal primitives. The only consequence is an increase of relevant constraints; e.g. MAX-L and MAX-M with identical weight are relevant instead of only MAX-•<sup>low</sup>.

## 4 The analysis: Activity asymmetries in ZAC

The HLT account in this paper is based on activity differences that 1) explain lexical contrasts between floating and non-floating tones, 2) keep track of underlyingly epenthetic tones that are different from non-epenthetic tones, 3) encode word boundaries as a purely phonological contrast and 4) predict different phonological behaviours for stems and phrases from predictably activity decay. The list in (19) summarizes the distribution of relevant phonological and morphological processes across morphological layers to ease the presentation of the theoretical account in the next subsections.

The activity decay for moras predicts an important first threshold: L-epenthesis applies at the stem layer but not at the phrase layer, simply because the moras are now too weak to justify tone epenthesis (19-a). They hence remain tone-less and are realized with a phonetic default M-tone.<sup>6</sup> For most words, tone morphemes are only added at the word (cf. the discussions in 2.1 and 6.1.2). Crucially, only tones with a default activity of 1 are associated at this point. Stronger tones  $T_{1.2}$  will remain floating until they are weakened enough to associate at the phrase layer (19-b). At the phrase layer, final H- and S-tones are predicted to spread and overwrite following originally epenthetic tones. This process only applies at the phrase layer since its context is not met at earlier layers: It is only triggered by a preference to avoid higher tones H and S that are followed by L-tones (19-c).

### (19) Distribution of processes across layers

|          | a. L-epenthesis | b. Association of tones  | c. H/S-Spread       |
|----------|-----------------|--------------------------|---------------------|
| Stem ①   | yes             | <i>(no context)</i>      | <i>(no context)</i> |
| Word ②   | yes             | Underlying / $T_1$ /     | <i>(no context)</i> |
| Phrase ③ | no              | Underlying / $T_{1.2}$ / | yes                 |

The formal implementation of these generalizations is given in the following. The predictable decay of tones and moras is discussed in subsection 4.1. Subsection 4.2 focusses on the processes at the stem and word layer: L-epenthesis and tone association of underlying tones  $T_1$ . Subsection 4.3 adds the processes at the phrase layer: Realization of the floating tones that started as strong / $T_{1.2}$ /, spreading of H and S, and the absence of L-epenthesis that results in tone-less TBU's at the phrase layer.

### 4.1 Activity adjustments across layers

The phonological grammar for ZAC predicts that all tones and all non-final moras loose 0.1 of their underlying activity at each optimization, schematized in (20) and (21). The general mechanism behind this activity decay was already introduced in subsection 3.1 where tableau (16) showed how a decay by 0.1 becomes optimal for every tone. Recall that tonal melodies are assumed to only be added at the word layer<sup>7</sup> and are hence only optimized twice, namely at the word (20-②) and phrase

<sup>6</sup>At the word layer, no relevant context provides tone-less TBU's, there is hence no way to know whether tone-less TBU's can still be repaired. The account presented here predicts that epenthesis still applies at the word layer but nothing hinges on this assumption.

<sup>7</sup>But cf. subsection 6.1.2 in the appendix that the account is also compatible with the view that some tones are already added at the stem layer.

layer (20-③). And a crucial assumption of the present account is the fact that some tones happen to leave the lexicon with a higher activity of 1.2, as is shown in (20-b) with the final tone.

(20) Activity loss across layers: Tones

|               | Underlying                           | Output stem ① | Output word ②                     | Output phrase ③                   |
|---------------|--------------------------------------|---------------|-----------------------------------|-----------------------------------|
| a. Morpheme 1 | /T <sub>1.0</sub> T <sub>1.0</sub> / | –             | T <sub>0.9</sub> T <sub>0.9</sub> | T <sub>0.8</sub> T <sub>0.8</sub> |
| b. Morpheme 2 | /T <sub>1.0</sub> T <sub>1.2</sub> / | –             | T <sub>0.9</sub> T <sub>1.1</sub> | T <sub>0.8</sub> T <sub>1.0</sub> |

The activity decay for moras (21) is not as symmetrical as the one for tones and excludes final moras that are preserved since they are in a prominent edge position. This simple phonological positional asymmetry will predict that word boundaries are retained as activity differences into the phrase layer without any reference to morphological boundaries. And in contrast to lexical tones that are only added at the word layer, moras are of course already present in the stem layer, schematized in (21).

(21) Activity decay across layers: Moras

|                                               | Underlying                                            | Output stem ①                                      | Output word ②                                       | Output phrase ③                                     |
|-----------------------------------------------|-------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| a. [[ [ μ μ μ ] <sub>St</sub> ] <sub>Wd</sub> | /μ <sub>1.0</sub> μ <sub>1.0</sub> μ <sub>1.0</sub> / | μ <sub>0.9</sub> μ <sub>0.9</sub> μ <sub>1.0</sub> | μ <sub>0.8</sub> μ <sub>0.8</sub> μ <sub>1.0</sub>  | μ <sub>0.7</sub> μ <sub>0.7</sub> μ <sub>1.0</sub>  |
| b. [[ [ μ μ ] <sub>St</sub> μ ] <sub>Wd</sub> | /μ <sub>1.0</sub> μ <sub>1.0</sub> /                  | μ <sub>0.9</sub> μ <sub>1.0</sub>                  | μ <sub>0.8</sub> μ <sub>0.9</sub> -μ <sub>1.0</sub> | μ <sub>0.7</sub> μ <sub>0.8</sub> -μ <sub>1.0</sub> |

The constraint interaction predicting this positionally asymmetric mora decay is shown in (23). The optimization is in principle absolutely parallel to the one for tone decay in (16): A context-free markedness constraint (22-a) demands weakening and thus excludes (23-a). Deletion of all moras as in (23-b), however, is not an option due to the inviolable weight of MAX-μ (22-b). Any possible activity adjustment is again restricted to 0.1 by IDENTACT-μ[≥0.1] (22-c) and makes (23-c) impossible. The only additional constraint that is necessary for mora decay is the positionally sensitive IDENTACT-μ# (22-e); a specific clone of the general IDENTACT-μ (22-d). The inviolably high weight of this faithfulness constraint excludes candidate (23-d) that symmetrically weakens all moras. The optimal candidate is hence (23-d) which weakens all non-final moras. That not only initial positions can be prominent and thus subject to designated faithfulness constraints is an extension of the classical positional faithfulness typology in Beckman (1997, 1998).

- (22)
- |    |                                                                                                                 |      |
|----|-----------------------------------------------------------------------------------------------------------------|------|
| a. | *μ                                                                                                              | W=10 |
|    | Assign -x violation for every mora μ <sub>x</sub> .                                                             |      |
| b. | MAX-μ                                                                                                           | W=∞  |
|    | Assign -x violation for every mora μ <sub>0</sub> corresponding to input μ <sub>x</sub> .                       |      |
| c. | IDENTACT-μ[≥0.1]                                                                                                | W=∞  |
|    | For every mora μ <sub>y</sub> corresponding to input μ <sub>x</sub> , assign -( x-y ) violation if ( x-y )>0.1. |      |
| d. | IDENTACT-μ                                                                                                      | W=1  |
|    | For every mora μ <sub>y</sub> corresponding to input μ <sub>x</sub> , assign -( x-y ) violation.                |      |
| e. | IDENTACT-μ#                                                                                                     | W=∞  |
|    | For every final mora μ <sub>y</sub> corresponding to input μ <sub>x</sub> , assign -( x-y ) violation.          |      |

(23) Activity loss for moras

| $\mu_1 \mu_1$            | IDENTACT- $\mu\#$<br>$\infty$ | IDENTACT- $\mu[\geq 0.1]$<br>$\infty$ | MAX- $\mu$<br>$\infty$ | * $\mu$<br>10 | IDENTACT- $\mu$<br>1 | $\mathcal{H}$ |
|--------------------------|-------------------------------|---------------------------------------|------------------------|---------------|----------------------|---------------|
| a. $\mu_1 \mu_1$         |                               |                                       |                        | -2            |                      | -20.0         |
| b.                       |                               |                                       | -2                     |               |                      | $-\infty$     |
| c. $\mu_{0.1} \mu_{0.1}$ | -0.9                          | -1.8                                  |                        | -0.2          | -1.8                 | $-\infty$     |
| d. $\mu_{0.9} \mu_{0.9}$ | -0.1                          |                                       |                        | -1.8          | -0.2                 | $-\infty$     |
| e. $\mu_{0.9} \mu_1$     |                               |                                       |                        | -1.9          | -0.1                 | -19.1         |

In the following subsections, the predictable activity decay for moras and tones is taken for granted and no candidate is considered that does not conform to the generalization that every tone and non-final mora (that is not deleted for independent reasons) decays by 0.1.

#### 4.2 Stem and word layer: Tone epenthesis and tonal melody association

Under the assumption that tonal melodies are only added at the word layer (cf. 6.1.2 in the appendix for a discussion), the optimization at the stem layer remains relatively straightforward: Only tone-less moras are optimized that are all provided with an epenthetic L-tone. The relevant constraints predicting tone epenthesis from their interaction are given in (24). A standard SPEC constraint demands that every TBU is provided with a tone; relativized to the activity of the TBU. The constraint DEP-T penalizes any inserted tone, again relativized to the activity of the inserted tone. In the absence of any constraint against weak output tones, this gradient constraint predicts that epenthesis is always minimal; a prediction that is crucial in ZAC and is discussed in more detail below. Finally, \*SPR penalizes tones associated to more than one TBU. Since M- and L-tones behave differently from H- and S-tones in their ability to spread (cf. (7)), there are different constraints penalizing the respective spreading processes; only \*SPR- $\bullet^{\text{low}}$  is relevant for now.

- (24)
- a. SPEC W=95  
Assign -x violation for every  $\text{TBU}_x$  that is not associated to a tonal root node  $\bullet$ .
  - b. DEP-T W=810  
Assign -x violation for every tonal root node  $\bullet_x$  without an input correspondent.
  - c. \*SPR- $\bullet^{\text{low}}$  W=100  
Assign -x violation for every [low] tonal root node  $\bullet_x^{\text{low}}$  that is associated to more than one TBU.

Tableau (25) optimizes a bimoraic stem. Realization of these two tone-less moras (25-a) induces -1.9 violations of SPEC: the combined activity of all moras not associated to a tone. Association to an epenthetic tone (25-b-d) avoids these violations, irrespective of how active this inserted tone is. That the default tone in ZAC is always L is taken to be ensured by an independent constraint interaction not shown here. The high weight of gradient DEP-T now ensures that candidate (25-b) with insertion of only a single minimally active  $L_{0.1}$ <sup>8</sup> that associates to all otherwise tone-less TBU's becomes optimal. This candidate violates DEP-T only by -0.1 and induces an additional -0.1 violation

<sup>8</sup>This implies that activity contrasts are restricted to not more than one decimal point here.



of both  $*\text{SPR-}\bullet^{\text{low}}$  and  $*\text{T}$  but the effect of these constraints on the harmony score is not as grave as the one of the -1.9 SPEC violations of (25-a) (cf. (26-i)). Insertion of a more active tone in candidate (25-c) that associates to all TBU's is harmonically bounded by (25-b) since it violates all constraints that (25-b) violates to a higher degree due to its high output activity. Insertion of two minimally active  $\text{L}_{0.1}$  tones in (25-d) avoids the -0.1 violation of  $*\text{SPR-}\bullet^{\text{low}}$  encountered by (25-b) but induces additional -0.1 violations of DEP-T, resulting in a worse harmony score.

To ease readability of the following tableaux, the output tone specification for each TBU is given in square brackets below the autosegmental representation (excluding any floating tones). These are not equivalent to a surface representations since they encode tone-less TBU's as "ø" which are – according to the assumptions made in this paper – phonetically realized as mid tones. Following the notation from above, word boundaries are given in square brackets '[]' but it is important to keep in mind that those are only intended to ease readability; they do not notate any prosodic boundary that the phonology has access to. I also want to emphasize that the constraint weights used in this account are just one exemplifying set of weights that are compatible with all weighting arguments relevant for ZAC. The weighting arguments illustrated in each optimization are given below each tableau in the following. Crucially, though, these only summarize the most essential constraint interactions since the tableaux do not contain the full set of constraints. I decided to move the full tableaux to the appendix (section 6) to allow the reader to focus on the crucial constraint interactions for a specific context.

(25) L-Epenthesis at ①

(cf. Appendix (71))

| $\mu_1$ $\mu_1$                                                                                                             | DEP-T<br>810 | $*\text{SPR-}\bullet^{\text{low}}$<br>100 | SPEC<br>95 | $*\text{T}$<br>8 | $\mathcal{H}$ |
|-----------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------|------------|------------------|---------------|
| a. $\mu_{0.9}$ $\mu_1$<br>[ø.ø]                                                                                             |              |                                           | -1.9       |                  | -180.5        |
| b. $\begin{array}{c} \text{L}_{0.1} \\ \swarrow \quad \searrow \\ \mu_{0.9} \quad \mu_1 \\ \text{[L.L]} \end{array}$        | -0.1         | -0.1                                      |            | -0.1             | -91.8         |
| c. $\begin{array}{c} \text{L}_1 \\ \swarrow \quad \searrow \\ \mu_{0.9} \quad \mu_1 \\ \text{[L.L]} \end{array}$            | -1           | -1                                        |            | -1               | -918          |
| d. $\begin{array}{c} \text{L}_{0.1} \quad \text{L}_{0.1} \\   \quad   \\ \mu_{0.9} \quad \mu_1 \\ \text{[L.L]} \end{array}$ | -0.2         |                                           |            | -0.2             | -163.6        |

(26) Weighting relations

|      |         |                                                   |     |                                                                                                  |
|------|---------|---------------------------------------------------|-----|--------------------------------------------------------------------------------------------------|
| i.   | $b > a$ | $1.9 \times \text{SPEC}$                          | $>$ | $0.1 \times \text{DEP-T} + 0.1 \times * \text{SPR-}\bullet^{\text{low}} + 0.1 \times * \text{T}$ |
| ii.  | $b > c$ | <i>harmonically bounded</i>                       |     |                                                                                                  |
| iii. | $b > d$ | $0.1 \times \text{DEP-T} + 0.1 \times * \text{T}$ | $>$ | $0.1 \times * \text{SPR-}\bullet^{\text{low}}$                                                   |

After the stem layer optimization, all TBU's of a stem are thus associated to the same weak  $L_{0.1}$ .<sup>9</sup> At the word layer, tonal morphemes are now added to stems. The possible underlying tonal specification of inflectional morphemes are given in (27). There are four assumptions that require some commenting, especially if one compares these underlying representations with the surface tonal melodies in (3). Firstly, there are lexical activity differences for underlying tones: The final tones of morpheme classes 2, 4, and 7 happen to have the higher underlying activity of 1.2. These highly active tones will be the 'floating' tones since they cannot associate at the word layer and have to wait until they are decayed enough at the phrase layer. Secondly, most of the TBU's realized as LS correspond only to an S-tone in (27). That these S-tones are always realized as an LS contour falls out from the inventory restrictions of ZAC that simply disallows any S-tones that are not preceded by an L-tone on any TBU. The account proposed here predicts that an epenthetic L is hence always inserted on the same TBU as the S. And thirdly, there are no floating L-tones in classes 3 and 13, contra the assumption in Villard (2015), discussed in more detail in subsection 6.1.1 in the appendix.

(27) Underlying representations: Tone melodies

| class            | class                | class             |
|------------------|----------------------|-------------------|
| 2. $L_1 S_{1.2}$ | 7. $L_1 M_1 S_{1.2}$ | 12. $M_1 M_1 L_1$ |
| 3. $L_1$         | 8. $L_1 H_1$         | 13. $M_1 S_1 L_1$ |
| 4. $M_1 H_{1.2}$ | 9. $L_1 L_1 S_1$     | 14. $M_1 H_1 M_1$ |
| 5. $M_1$         | 10. $M_1 M_1$        | 15. $L_1 M_1 L_1$ |
| 6. $L_1 M_1$     | 11. $M_1 H_1$        |                   |

The following tableaux show the association of these tonal melodies at the word layer. Tableaux (29), (31), and (33) illustrate how the tonal morpheme  $M_1 H_1$  (class 11) associates to stems of different lengths before the contrasting behaviour of morpheme  $M_1 H_{1.2}$  (class 4) containing a strongly active tone is shown in tableau (36). The additional constraints that become relevant for these contexts are given in (28). \*FLT-• (28-a) is the gradient version of a standard constraint against tones without any association and hence favours association of all tones added at the word layer. Since all TBU's are associated to a  $L_{0.1}$  when they enter the word layer, all tonal association at the word layer necessarily involves tonal overwriting, the faithfulness constraint against tonal deletion  $MAX-•^{low}$  (28-b) and  $MAX-•^{high}$  (28-c) become relevant as well as  $MAX-AL^{low}$  (28-b), a faithfulness constraint penalizing any deassociation of association lines linking tones with a low register. That the corresponding  $MAX-AL^{high}$  has a higher weight in ZAC is discussed below. Both these constraint families hence encode a difference that is central for the account: [low] register tonal root nodes are easier to deassociate and delete whereas [high] register tonal root nodes are impossible to delete and deassociate once they found their TBU.

- (28) a. \*FLT-• W=379  
Assign -x violation for every tonal root node  $•_x$  that is not associated to a TBU.
- b.  $MAX-•^{low}$  W=115  
Assign -x violation for every input [low] tonal root node  $•_x$  without an output corre-

<sup>9</sup>I thus omit DEP-T in the following tableaux. Insertion of another epenthetic tone will only become a potentially helpful strategy again in long overwriting cases in tableau (57). Similarly, \*T is omitted as well since it does not contribute to the crucial weighting relations in most contexts due to its low weight. I also omit \*SPR-•<sup>low</sup> for now. Realization of floating tonal melodies at the word layer always reduces the number of \*SPR-•<sup>low</sup> violations since the originally epenthetic  $L_{0.1}$  will deassociate from TBU's; the constraint hence rather favors the winner in the following tableaux.

spondent.

- c.  $\text{MAX-}\bullet^{\text{high}}$   $W=\infty$   
Assign -x violation for every input [high] tonal root node  $\bullet_x$  without an output correspondent.
- d.  $\text{MAX-AL}^{\text{low}}$   $W=320$   
Assign -x violation for every association line linking a [low] tonal root node  $\bullet_x$  to a TBU in the input if that association line has no output correspondent.

Tableau (29) starts with a trimoraic base, hence one where there are more TBU's than tones. Leaving both tones unassociated as in candidate (29-a) violates  $\ast\text{FLT-}\bullet$  by 1.8 – the combined output activity of all tones not linked to a TBU. To ease readability, unassociated tones are marked with a circle in the following tableaux. Deleting those tones in (29-b) avoids any violation of  $\ast\text{FLT-}\bullet$  but induces a -1 violation of both  $\text{MAX-}\bullet^{\text{low}}$  and  $\text{MAX-}\bullet^{\text{high}}$ . That both have different weights in ZAC and deletion of H/S-tone is inherently worse than deletion of a L/M-one is discussed in more detail below. Association of both tones to a TBU in candidate (29-c), on the other hand, only induces violations of  $\text{MAX-AL}^{\text{low}}$ . Since the activity of an association line is always taken to equal the activity of the higher-tier element, the constraint is only violated by -0.2 since two 0.1-association lines remain unrealized. The weighting arguments in (30) show that the low weight of  $\text{MAX-AL}^{\text{low}}$  straightforwardly predicts this candidate as optimal.

Note that none of the following tableaux considers a candidate that creates illicit tonal specifications on TBU's, most notably S-tones that do not share a TBU with an L and contours other than LH or LS. These specifications are taken to be excluded by unviolable inventory constraints. Tableau (29) shows that the rule-based generalization that tone association always starts at the right edge is an epiphenomenon in this HLT model: It falls out from the fact that stems are already specified for an epenthetic tone and tonal morphemes are suffixed. The alternative tonal association  $\ast[\text{M.H.L}]$  for (29) is hence harmonically bounded since it involves complete deletion of one  $L_{0.1}$  and insertion of another.

(29) Tonal melody association at ②: More TBU's than tones

(cf. Appendix (72))

| <div style="text-align: center;"> <math>L_{0.1}</math>    <math>M_1</math>    <math>H_1</math><br/> <math>\mu_{0.9}</math> <math>\mu_{0.9}</math> <math>\mu_1</math> </div>                                                  | $\text{MAX-}\bullet^{\text{high}}$ | $\ast\text{FLT-}\bullet$ | $\text{MAX-AL}^{\text{low}}$ | $\text{MAX-}\bullet^{\text{low}}$ | $\mathcal{H}$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------|------------------------------|-----------------------------------|---------------|
| <div style="text-align: center;"> a. <math>L_{0.1}</math>    <math>\textcircled{M_{0.9}}</math> <math>\textcircled{H_{0.9}}</math><br/> <math>\mu_{0.8}</math> <math>\mu_{0.8}</math> <math>\mu_1</math><br/> [L.L.L] </div> |                                    | -1.8                     |                              |                                   | -682.2        |
| <div style="text-align: center;"> b. <math>L_{0.1}</math><br/> <math>\mu_{0.8}</math> <math>\mu_{0.8}</math> <math>\mu_1</math><br/> [L.L.L] </div>                                                                          | -1                                 |                          |                              | -1                                | $-\infty$     |
| <div style="text-align: center;"> c. <math>L_{0.1}</math> <math>M_{0.9}</math> <math>H_{0.9}</math><br/> <math>\mu_{0.8}</math> <math>\mu_{0.8}</math> <math>\mu_1</math><br/> [L.M.H] </div>                                |                                    |                          | -0.2                         |                                   | -64.0         |

(30) Weighting relations

|     |         |                                                            |     |                           |
|-----|---------|------------------------------------------------------------|-----|---------------------------|
| i.  | $c > a$ | $1.8 \times *FLT-\bullet$                                  | $>$ | $0.2 \times MAX-AL^{low}$ |
| ii. | $c > b$ | $1 \times MAX-\bullet^{high} + 1 \times MAX-\bullet^{low}$ | $>$ | $0.2 \times MAX-AL^{low}$ |

Now we turn to a bimoraic base combined with the same tonal morpheme; there are hence the same number of tones and TBU's. As in (30), both floating tones are predicted to associate due to the high weight of  $*FLT-\bullet$  and  $MAX-\bullet^{high}$ , excluding (31-a) and in (31-b). However, in this context, the price for realizing the tones is higher since the originally epenthetic  $L_{0.1}$  now lacks a TBU. Candidates (31-c) and (31-d) compare the two possible options for this tone: It could remain floating and induce a -0.1 violation of  $*FLT-\bullet$  (31-c) or it could delete altogether inducing a -0.1 violation of  $MAX-\bullet^{low}$ . Since  $MAX-\bullet^{low}$  has a lower weight than  $*FLT-\bullet$  (32-iii), the latter strategy is predicted to be optimal.

That epenthesis is always minimal  $L_{0.1}$  insertion hence predicts an inherent asymmetry between underlyingly present and originally epenthetic tones without any reference to morphological affiliation and across different morphological layers. Underlying tones will always easily overwrite originally epenthetic  $L_{0.1}$  tones simply because the latter violate  $MAX-\bullet^{low}$  to a lower degree when left unrealized. For all following tableaux, it is taken for granted that  $L_{0.1}$  is deleted to make room for an otherwise more active floating tone.

(31) Tone association at ②: Same number of floating tones and TBU's (cf. Appendix (73))

|    | $L_{0.1}$ $M_1$ $H_1$<br>$\mu_{0.9}$ $\mu_1$                  | $MAX-\bullet^{high}$<br>$\infty$ | $*FLT-\bullet$<br>379 | $MAX-AL^{low}$<br>320 | $MAX-\bullet^{low}$<br>115 | $\mathcal{H}$ |
|----|---------------------------------------------------------------|----------------------------------|-----------------------|-----------------------|----------------------------|---------------|
| a. | $L_{0.1}$ $M_{0.9}$ $H_{0.9}$<br>$\mu_{0.8}$ $\mu_1$<br>[L.L] |                                  | -1.8                  |                       |                            | -682.2        |
| b. | $L_{0.1}$ $M_{0.9}$ $H_{0.9}$<br>$\mu_{0.8}$ $\mu_1$<br>[L.L] | -1                               |                       |                       | -1                         | $-\infty$     |
| c. | $L_{0.1}$ $M_{0.9}$ $H_{0.9}$<br>$\mu_{0.8}$ $\mu_1$<br>[M.H] |                                  | -0.1                  | -0.2                  |                            | -101.9        |
| d. | $M_{0.9}$ $H_{0.9}$<br>$\mu_{0.8}$ $\mu_1$<br>[M.H]           |                                  |                       | -0.2                  | -0.1                       | -75.5         |

(32) Weighting relations

|      |         |                                                              |     |                                                          |
|------|---------|--------------------------------------------------------------|-----|----------------------------------------------------------|
| i.   | $d > a$ | $1.8 \times *FLT-\bullet$                                    | $>$ | $0.2 \times MAX-AL^{low} + 0.1 \times MAX-\bullet^{low}$ |
| ii.  | $d > b$ | $1 \times MAX-\bullet^{high} + 0.9 \times MAX-\bullet^{low}$ | $>$ | $0.2 \times MAX-AL^{low}$                                |
| iii. | $d > c$ | $0.1 \times *FLT-\bullet$                                    | $>$ | $0.1 \times MAX-\bullet^{low}$                           |

If the floating tonal melody added at the word layer contains more tones than TBU's are present in

the stem, not all tones can find a host, even with  $L_{0.1}$  being deleted. This is shown in (33) where the  $/M_1 H_1/$  tonal morpheme is combined with a monomoraic stem. The weighting relations already introduced above make it clear that (33-a) is sub-optimal: Deletion of  $L_{0.1}$  is better than leaving a  $T_{0.9}$  floating. But since there are two  $T_1$ 's in the input, two additional choices need to be made: Is the first  $T_1$  realized (33-b) or the second (33-c+d) and is the tone that remains without a host left floating (33-b+c) or deleted (33-d)? The optimal strategy in ZAC is predicted by the relative weighting difference between  $*FLT-\bullet$  and lower-weighted  $MAX-\bullet^{low}$ : Deletion of even a fully active input [low] tonal root node is always better than leaving any tone floating. Since the [low] tone in (33) is the initial tone of the tonal melody, it is hence predicted that the final tone of the tonal melody associates and both the originally epenthetic  $L_{0.1}$  and the 'left-over'  $M_{0.9}$  are deleted as in candidate (33-d). This is the better strategy than leaving any tone floating (33-b+c) and it definitely is better than deleting the final H-tone and only associating the M-tone. This candidate is not even shown in (33) since it is excluded by  $MAX-\bullet^{high}$  and its inviolable weight.

(33) Tone association at ②: Fewer TBU's than tones

(cf. Appendix (74))

| $L_{0.1} M_1 H_1$<br>$\mu_1$                     | $\infty$<br>$MAX-\bullet^{high}$ | 379<br>$*FLT-\bullet$ | 320<br>$MAX-AL^{low}$ | 115<br>$MAX-\bullet^{low}$ | $\mathcal{H}$ |
|--------------------------------------------------|----------------------------------|-----------------------|-----------------------|----------------------------|---------------|
| a. $L_{0.1} (M_{0.9} H_{0.9})$<br>$\mu_1$<br>[L] |                                  | -1.8                  |                       |                            | -682.2        |
| b. $M_{0.9} (H_{0.9})$<br>$\mu_1$<br>[M]         |                                  | -0.9                  | -0.1                  | -0.1                       | -384.6        |
| c. $(M_{0.9} H_{0.9})$<br>$\mu_1$<br>[H]         |                                  | -0.9                  | -0.1                  | -0.1                       | -384.6        |
| d. $H_{0.9}$<br>$\mu_1$<br>[H]                   |                                  |                       | -0.1                  | -1.1                       | -158.5        |

(34) Weighting relations

|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| i. $d > a$    | $1.8 \times *FLT-\bullet > 0.1 \times MAX-AL^{low} + 1.1 \times MAX-\bullet^{low}$ |
| ii. $d > b/c$ | $0.9 \times *FLT-\bullet > 1 \times MAX-\bullet^{low}$                             |

The choice in favor of the final tone in (33) followed since the tonal melody 11 contains a [low] tone followed by a [high] tone. Interestingly, this explanation can be generalized to all other contexts as well and already predicts the rule-based generalization that tonal association starts from the right and associates the rightmost tone with the rightmost TBU. It follows from the inventory restrictions of ZAC, namely the fact that very few tonal melodies have a higher tone preceding a lower tone.<sup>10</sup>

<sup>10</sup>The only exceptions are classes 13 M.LS.L and 14 M.H.M. However, both these melodies are never attested for

Now we can finally turn to the strong underlying tones, namely the final  $H_{1,2}$  and  $S_{1,2}$  of classes 2, 4, and 7. Tableau (36) optimizes a class 4 morpheme since it nicely contrasts with the class 11 morpheme optimized above: Both tonal melodies are identical in terms of their tonal content but differ in the activity of their final tone. The additional constraint necessary in these contexts is (35), a markedness constraint penalizing any associated tone with an activity higher than one. It only penalizes strong tones that are associated since it only cares about phonetically realized tones that are integrated into the overall structure and ignores any floating tones. Since this constraint is never violated in an optimal structure, it again receives the ‘inviolable’ weight of  $\infty$  below.

- (35) \*STRONG-• W= $\infty$   
Assign -x violation for every tonal root node •<sub>x</sub> if  $x > 1$  that is associated to a TBU.

According to the predictable activity decay established in tableau (16), an underlying  $H_{1,2}$  surfaces with 1.1 activity in the output. Importantly, the  $\infty$  weight of IDENTACT-T[ $\geq 0.1$ ] (cf. again (16)) excludes more reduction of this activity – if the tone is not deleted, it hence has an activity higher than 1. The first difference between tableau (36) and the corresponding optimization in (29) is thus the fact that \*FLT-• is violated more if  $H_{1,1}$  remains floating as in candidates (36-a) and (36-c). Association of  $H_{1,1}$  to a TBU as in candidate (36-b), on the other hand, induces a fatal violation of \*STRONG-• – the strong tone can hence not associate. The choice is now to either leave it floating (36-c) or to delete it (36-d) since both options avoid the fatal violation of \*STRONG-•. Since -1.2 violations of MAX-•<sup>high</sup> are worse than -1.1 violations of \*FLT-• (37-iii), the former strategy emerges as optimal.

- (36) Tone association at ②: Strong tones remain floating (cf. Appendix (75))

|    | $L_{0,1}$ $M_1$ $H_{1,2}$<br>$\mu_{0,9}$ $\mu_{0,9}$ $\mu_1$                | *STRONG-•<br>$\infty$ | MAX-• <sup>high</sup><br>$\infty$ | *FLT-•<br>379 | MAX-AL <sup>low</sup><br>320 | MAX-• <sup>low</sup><br>115 | $\mathcal{H}$ |
|----|-----------------------------------------------------------------------------|-----------------------|-----------------------------------|---------------|------------------------------|-----------------------------|---------------|
| a. | $L_{0,1}$ $M_{0,9}$ $H_{1,1}$<br>$\mu_{0,8}$ $\mu_{0,8}$ $\mu_1$<br>[L.L.L] |                       |                                   | -2            |                              |                             | -758          |
| b. | $L_{0,1}$ $M_{0,9}$ $H_{1,1}$<br>$\mu_{0,8}$ $\mu_{0,8}$ $\mu_1$<br>[L.M.H] | -1                    |                                   |               | -0.2                         |                             | $-\infty$     |
| c. | $L_{0,1}$ $M_{0,9}$ $H_{1,1}$<br>$\mu_{0,8}$ $\mu_{0,8}$ $\mu_1$<br>[L.L.M] |                       |                                   | -1.1          | -0.1                         |                             | -448.9        |
| d. | $L_{0,1}$ $M_{0,9}$<br>$\mu_{0,8}$ $\mu_{0,8}$ $\mu_1$<br>[L.L.M]           |                       | -1.2                              |               | -0.1                         |                             | $-\infty$     |

bases shorter than two moras – there is hence never the question of leaving a higher tone unassociated. The only additional assumption necessary to predict all tonal associations correctly is a positional faithfulness constraint MAX-T# that protects a final tone in case the contrast between MAX-•<sup>high</sup> and MAX-•<sup>low</sup> can’t decide which floating tone to associate in case there are not enough TBU’s.

(37) Weighting relations

|      |         |                                                        |     |                           |
|------|---------|--------------------------------------------------------|-----|---------------------------|
| i.   | $c > a$ | $0.9 \times *FLT-\bullet$                              | $>$ | $0.1 \times MAX-AL^{low}$ |
| ii.  | $c > b$ | $1.1 \times *STRONG-\bullet + 0.1 \times MAX-AL^{low}$ | $>$ | $1.1 \times *FLT-\bullet$ |
| iii. | $c > d$ | $1.2 \times MAX-\bullet^{high}$                        | $>$ | $1.1 \times *FLT-\bullet$ |

Importantly, underlying  $H_{1.2}$  and  $S_{1.2}$  hence remain floating in the output of the stem layer and are transferred into the phrase layer. This contrasts with input  $M_1$  and  $L_1$  that remain without an output host in contexts where there are too few TBU's: Those are deleted and hence never make it to the phrase layer. This was shown in tableau (33); the relevant weighting relations are repeated again in (38).

(38) The fate of tones that can't associate at ②

- a. Deletion of underlying  $M_1/L_1$ , cf. (34)-ii  
 $0.9 \times *FLT-\bullet > 1 \times MAX-\bullet^{low}$
- b. Floating of underlying  $H_{1.2}/S_{1.2}$ , cf. (37)-iii  
 $1.2 \times MAX-\bullet^{high} > 1.1 \times *FLT-\bullet$

### 4.3 Phrase layer: Tonal spreading and floating tone association

In this subsection we now turn to the next layer ③, namely the phrase layer where the words that were optimized at layer ② are combined. The first process we discuss is the spreading of H and S. In subsection 2.2, it was generalized that both H and S spread to 'X'-TBU's, namely those that are not associated to a tone of a tonal morpheme (27). Since the present account is based on the assumption that all tone-less TBU's are provided with an epenthetic tone at the stem layer, these are hence associated to a  $L_{0.1}$  tone at the phrase layer. Spreading in my account is hence not triggered by the need to fill tone-less TBU's with tones, it rather applies to avoid sequences of higher tones followed by a low tone, ensured by the markedness constraint  $*hL$  (39). The dispreference for H-toned TBU's followed by L-toned ones is reminiscent of the inventory restriction discussed above for tonal morphemes: Nearly all of them have the higher tones H and S as the final one of the melody and there is in fact only a single sequence of a high register tone followed by an L-tone (class 13). Since X-TBU's are all associated to an  $L_{0.1}$ -tone at the phrase layer, any preceding word ending in a H- or S-tone will violate  $*hL$ . The optimal repair to avoid this marked sequence is the deletion of any following  $L_{0.1}$ -tones and spreading of the H and S to all otherwise tone-less TBU's. A constraint that will thus be violated in the optimal candidates is  $*SPR-\bullet^{high}$  (39-b), the counterpart to  $*SPR-\bullet^{low}$ . To ensure that an offending HL or SL sequence is repaired by overwriting of the  $L_{0.1}$  and not by de-association of the H/S-tone, we also need to consider  $MAX-AL^{high}$  (39-c), the higher-weighted counterpart of  $MAX-AL^{low}$ . Since it is never violated in ZAC, it again gets the inviolable weight of  $\infty$ .

- (39) a.  $*hL$  W=152  
Assign  $-(n \times x)$  violations for every TBU associated to a [high] tonal root node  $^h\bullet_x$  that is immediately followed by a sequence of n [low,LOW] tonal root nodes  $^{lL}\bullet$ .
- b.  $*SPR-\bullet^{high}$  W=75  
Assign -x violation for every [high] tonal root node  $^l\bullet_x$  that is associated to more than

one TBU.

c. MAX-AL<sup>high</sup>

W=∞

Assign -x violation for every association line linking a [high] tonal root node •<sub>x</sub> to a TBU in the input if that association line has no output correspondent.

Tableau (40) optimizes the first context where \*hL becomes relevant: A monomoraic word associated to H<sub>0.9</sub> is combined with a following bimoraic word associated to L<sub>0.1</sub> and M<sub>0.9</sub> in the output of the word layer. Faithful realization of all three tones in candidate (40-a) violates \*hL by -0.8 since an output-H<sub>0.8</sub> precedes one L-tone. This violation is avoided by deletion of L<sub>0.1</sub> in candidate (40-b) which violates MAX-AL<sup>low</sup> and MAX-•<sup>low</sup> only by -0.1 but also induces a -0.7 violation of SPEC since a word-medial TBU remains without any tone.<sup>11</sup> Spreading of the H in candidate (40-c) avoids this SPEC violation and trades it for a -0.8 violation of lower-weighted \*SPR-•<sup>high</sup>. This candidate emerges as optimal in this context. For completeness, candidate (40-d) avoids the \*hL violation by deassociation of the H-tone and spreading of the L-tone to avoid a tone-less TBU. This candidate is already excluded by its -0.9 violations of inviolable MAX-AL<sup>high</sup> but would of course also induce a -0.8 violation of \*FLT-•. This tableau already explains why only H<sub>0.9</sub> and S<sub>0.9</sub> ever spread in ZAC: There simply is no reason for M<sub>0.9</sub> and L<sub>0.9</sub> to ever spread to a X-TBU (=overwrite a L<sub>0.1</sub>) since the trigger for spreading is only \*hL. To ease readability of the following phrase layer tableaux, the first line of the input repeats the notation used above with ‘X’ and adds the tonal classes of the morphemes as superscripts.

(40) H-Spreading at ③: Overwriting of L<sub>0.1</sub>

(cf. Appendix (76))

| /H/ <sup>11</sup> /X.M/ <sup>5</sup> |                                                                                                                     | MAX-AL <sup>high</sup> | MAX-AL <sup>low</sup> | *hL  | MAX-• <sup>low</sup> | *SPR-• <sup>low</sup> | SPEC | *SPR-• <sup>high</sup> | ℋ      |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------|------|----------------------|-----------------------|------|------------------------|--------|
| H <sub>0.9</sub>                     | L <sub>0.1</sub> M <sub>0.8</sub>                                                                                   | ∞                      | 320                   | 152  | 115                  | 100                   | 95   | 75                     |        |
| μ <sub>1</sub>                       | μ <sub>0.7</sub> μ <sub>1</sub>                                                                                     |                        |                       |      |                      |                       |      |                        |        |
| a.                                   | H <sub>0.8</sub> L <sub>0.1</sub> M <sub>0.8</sub><br>μ <sub>0.9</sub> μ <sub>0.7</sub> μ <sub>1</sub><br>[H] [L.M] |                        |                       | -0.8 |                      |                       |      |                        | -121.6 |
| b.                                   | H <sub>0.8</sub> M <sub>0.8</sub><br>μ <sub>0.9</sub> μ <sub>0.7</sub> μ <sub>1</sub><br>[H] [ø.M]                  |                        | -0.1                  |      | -0.1                 |                       | -0.7 |                        | -110.0 |
| c.                                   | H <sub>0.8</sub> M <sub>0.8</sub><br>μ <sub>0.9</sub> μ <sub>0.7</sub> μ <sub>1</sub><br>[H] [H.M]                  |                        | -0.1                  |      | -0.1                 |                       |      | -0.8                   | -103.5 |
| d.                                   | H <sub>0.8</sub> L <sub>0.1</sub> M <sub>0.8</sub><br>μ <sub>0.9</sub> μ <sub>0.7</sub> μ <sub>1</sub><br>[L] [L.M] | -0.9                   |                       |      |                      | -0.1                  |      |                        | -∞     |

(41) Weighting relations

<sup>11</sup>This candidate also violates another constraint that is taken to be unviolable in ZAC: \*hø, introduced in (54). Since this constraint is not yet crucial, I decided to introduce it later. But cf. the appendix for the full tableaux with all constraints.



|      |         |                                                            |     |                                                                                           |
|------|---------|------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------|
| i.   | $c > a$ | $0.8 \times *hL$                                           | $>$ | $0.1 \times MAX-AL^{low} + 0.1 \times MAX-\bullet^{low} + 0.7 \times *SPR-\bullet^{high}$ |
| ii.  | $c > b$ | $0.7 \times SPEC$                                          | $>$ | $0.8 \times *SPR-\bullet^{high}$                                                          |
| iii. | $c > d$ | $0.9 \times MAX-AL^{high} + 0.1 \times *SPR-\bullet^{low}$ | $>$ | $0.1 \times MAX-AL^{low} + 0.1 \times MAX-\bullet^{low} + 0.8 \times *SPR-\bullet^{high}$ |

Crucially, only originally epenthetic and hence weak  $L_{0.1}$  tones can ever disassociate in order to satisfy  $*hL$ , originally underlying and hence stronger  $L_{0.9}$  tones have to be preserved. Some  $*hL$  violations are hence unavoidable in ZAC. This is shown in (42) where a H-toned monomoraic word is now followed by a monomoraic L-toned word. As in (41-a), realization of both induces a -0.8 violation of  $*hL$  but the spreading repair in (42-c) fares worse since it induces -0.9 violations of both  $MAX-AL^{low}$  and  $MAX-\bullet^{low}$ .

(42) No H-Spreading at ③: No overwriting of  $L_{0.8}$  (cf. Appendix (77))

| $/H/^{11} /L/^{13}$ |                                                               | $MAX-AL^{high}$ | $MAX-AL^{low}$ | $*hL$ | $MAX-\bullet^{low}$ | $*SPR-\bullet^{low}$ | $SPEC$ | $*SPR-\bullet^{high}$ |               |
|---------------------|---------------------------------------------------------------|-----------------|----------------|-------|---------------------|----------------------|--------|-----------------------|---------------|
| $H_{0.9}$           | $L_{0.9}$                                                     | $\infty$        | 320            | 152   | 115                 | 100                  | 95     | 75                    | $\mathcal{H}$ |
| $\mu_1$             | $\mu_1$                                                       |                 |                |       |                     |                      |        |                       |               |
|                     |                                                               |                 |                |       |                     |                      |        |                       |               |
| a.                  | $H_{0.8} \quad L_{0.8}$<br>$\mu_{0.9} \quad \mu_1$<br>[H] [L] |                 |                | -0.8  |                     |                      |        |                       | -121.6        |
| b.                  | $H_{0.8}$<br>$\mu_{0.9} \quad \mu_1$<br>[H] [H]               |                 | -0.9           |       | -0.9                |                      |        | -0.8                  | -451.5        |

(43) Weighting relation

|         |                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------|
| $a > b$ | $0.9 \times MAX-AL^{low} + 0.9 \times MAX-\bullet^{low} + 0.8 \times *SPR-\bullet^{high} > 0.8 \times *hL$ |
|---------|------------------------------------------------------------------------------------------------------------|

The threshold that only weak and hence originally epenthetic tones can be overwritten but not strong and thus originally underlying ones is summarized in (44). It is crucially only possible in a system with gradient activity.

(44) Overwriting to avoid violations of  $*hL$  at ②?

- Overwriting of originally epenthetic  $L_{0.1}$ , cf. (41)-i  
 $0.8 \times *hL > 0.1 \times MAX-AL^{low} + 0.1 \times MAX-\bullet^{low} + 0.8 \times *SPR-\bullet^{high}$
- No overwriting of originally underlying  $L_{0.9}$ , cf. (43)  
 $0.9 \times MAX-AL^{low} + 0.9 \times MAX-\bullet^{low} + 1 \times SPEC > 0.8 \times *hL$

For completeness, tableau (45) illustrates that the general weighting relations established above also predict long-distance spreading across potentially many word boundaries. It optimizes a context where a H-final word is followed by a bimoraic class 1 word and another bimoraic class 3 word. The H is thus followed by four L-toned TBU's, three are associated to a weak  $L_{0.1}$  and one to a strong  $L_{0.8}$ . Realization of these underlying tone specifications in (45-a) hence violates  $*hL$  by -3.2: The activity of the H-tone (=0.8) taken times the number of following L-tones (=4). Deletion of the two weak  $L_{0.1}$ 's in (45-b) allows to get rid of -2.4 of the  $*hL$  violations. As was already established in (42), every -0.8 violation of  $*hL$  caused by a following  $L_{0.8}$  must be tolerated and can't be repaired.

(cf. Appendix (78))

| $/H/^{11} /X.X/^{1} /X.L/^{3}$ |                        | $\infty$ | MAX-AL <sup>high</sup> | MAX-AL <sup>low</sup> | *hL  | MAX- <sup>low</sup> | *SPR- <sup>low</sup> | SPEC | *SPR- <sup>high</sup> | $\mathcal{H}$ |
|--------------------------------|------------------------|----------|------------------------|-----------------------|------|---------------------|----------------------|------|-----------------------|---------------|
|                                |                        |          |                        |                       |      |                     |                      |      |                       |               |
| a.                             | <p>[H] [L.L] [L.L]</p> |          |                        | -3.2                  |      | -0.1                |                      |      |                       | -496.4        |
| b.                             | <p>[H] [H.H] [H.L]</p> |          | -0.3                   | -0.8                  | -0.2 |                     |                      | -2.4 |                       | -420.6        |

(46) Weighting relations

|    |         |                                                                                                                                      |
|----|---------|--------------------------------------------------------------------------------------------------------------------------------------|
| i. | $b > a$ | $2.4 \times *hL + 0.1 \times *SPR \cdot^{low} > 0.3 \times MAX-AL^{low} + 0.2 \times MAX \cdot^{low} + 2.4 \times *SPR \cdot^{high}$ |
|----|---------|--------------------------------------------------------------------------------------------------------------------------------------|

The second important process at the phrase layer is the association of floating tones. The originally strongly active  $H_{1.2}$  and  $S_{1.2}$  that had to remain floating at the word layer are now finally weakened enough to be associated. Similarly to the pressure to avoid HL sequences, their need to be associated is not important enough to de-associate tones with more activity than 0.1. Only originally epenthetic  $L_{0.1}$  tones are thus ever deleted to allow realization of floating tones at ③. Tableau (47) shows the phrase layer optimization of a class 4 word with a floating  $H_{1.1}$  in the input followed by a trimoraic class 10 word that starts with a X/ $L_{0.1}$ -TBU. Leaving the H-tone unassociated (47-a) induces a -1 violation of \*FLT-• whereas deletion of the tone avoids this violation but adds a -1.1 violation of MAX-•<sup>high</sup>. Realization of the H-tone on the following TBU implies only -0.1 violations of both MAX-AL<sup>low</sup> and MAX-•<sup>low</sup> and emerges as optimal. Crucially, predictable activity decay ensures that the tone only has 1 activity left in the output and can be realized without any violation of \*STRONG-•.

(47) Floating tone overwrites  $L_{0.1}$  at ③

(cf. Appendix (79))

| $/M(H)^4/X.M.M.^{10}$<br>$\begin{array}{cccc} M_{0.9} & H_{1.1} & L_{0.1} & M_{0.9}M_{0.9} \\ \mu_1 & & \mu_{0.8} & \mu_{0.8} \mu_1 \end{array}$              | $M_{AX-\bullet}^{high}$<br>$\infty$ | $*F_{IT-\bullet}$<br>379 | $M_{AX-AL}^{low}$<br>320 | $M_{AX-\bullet}^{low}$<br>115 | $\mathcal{H}$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------|--------------------------|-------------------------------|---------------|
| a. $\begin{array}{cccc} M_{0.8} & \textcircled{H_1} & L_{0.1} & M_{0.8}M_{0.8} \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} \mu_1 \\ [M] & [L.M.M] & & \end{array}$ |                                     | -1                       |                          |                               | -379.0        |
| b. $\begin{array}{cccc} M_{0.8} & & L_{0.1} & M_{0.8}M_{0.8} \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} \mu_1 \\ [M] & [L.M.M] & & \end{array}$                   | -1.1                                |                          |                          |                               | $-\infty$     |
| c. $\begin{array}{cccc} M_{0.8} & H_1 & & M_{0.8}M_{0.8} \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} \mu_1 \\ [M] & [H.M.M] & & \end{array}$                       |                                     |                          | -0.1                     | -0.1                          | -43.5         |

(48) Weighting relations

|     |         |                                 |     |                                                          |
|-----|---------|---------------------------------|-----|----------------------------------------------------------|
| i.  | $c > a$ | $1 \times *FLT-\bullet$         | $>$ | $0.1 \times MAX-\bullet^{low} + 0.1 \times MAX-AL^{low}$ |
| ii. | $c > a$ | $1.1 \times MAX-\bullet^{high}$ | $>$ | $0.1 \times MAX-\bullet^{low} + 0.1 \times MAX-AL^{low}$ |

Given the discussion above that \*hL causes repairs to avoid sequences of H/S followed by L, one might wonder whether realization of an otherwise floating H-tone is indeed important enough to create new \*hL violations. This configuration arises if a floating H or S precedes a word that starts with a  $L_{0.1}L_{0.9}$  sequence. As tableau (49) briefly shows, the weighting relations correctly predict that a floating tone is realized in these contexts (49-b) even though this implies an additional \*hL violation. Even the combined effect of \*hL and -0.1 violations of  $MAX-AL^{low}$  and  $MAX-\bullet^{low}$  is still not enough to overcome a full violation of \*FLT-• in (49-a)

(49) Floating tone overwrites  $L_{0.1}$  at ③, despite \*hL (cf. Appendix (80))

| $/M(H)^4/X.L.M^6$                                                                                                                      |                                                                                                                                                                               | $MAX-\bullet^{high}$ | $*FLT-\bullet$ | $MAX-AL^{low}$ | *hL | $MAX-\bullet^{low}$ | $\mathcal{H}$ |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|----------------|-----|---------------------|---------------|
| $  \begin{array}{ccccccc}  M_{0.9} & H_{1.1} & L_{0.1} & L_{0.9} & M_{0.9} \\  \mu_1 & & \mu_{0.8} & \mu_{0.8} & \mu_1  \end{array}  $ |                                                                                                                                                                               | $\infty$             | 379            | 320            | 152 | 115                 |               |
| a.                                                                                                                                     | $  \begin{array}{ccccccc}  M_{0.8} & \textcircled{H_1} & L_{0.1} & L_{0.8} & M_{0.8} \\  \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} & \mu_1 \\  & & [M] & [L.L.M] &   \end{array}  $ |                      | -1             |                |     |                     | -379.0        |
| b.                                                                                                                                     | $  \begin{array}{ccccccc}  M_{0.8} & H_1 & & L_{0.8} & M_{0.8} \\  \mu_{0.9} & & & \mu_{0.7} & \mu_{0.7} & \mu_1 \\  & & & [M] & [H.L.M] &   \end{array}  $                   |                      |                | -0.1           | -1  | -0.1                | -195.5        |

(50) Weighting relations

|    |         |                         |     |                                                                         |
|----|---------|-------------------------|-----|-------------------------------------------------------------------------|
| i. | $b > a$ | $1 \times *FLT-\bullet$ | $>$ | $0.1 \times MAX-AL^{low} + 1 \times *hL + 0.1 \times MAX-\bullet^{low}$ |
|----|---------|-------------------------|-----|-------------------------------------------------------------------------|

We hence have established that floating H- and S-tones in ZAC overwrite  $L_{0.1}$ -tones in order to get realized. If they are immediately followed by a TBU associated to an originally underlying tone  $T_{0.9}$ , however, they have no chance of getting realized and remain floating. This is shown in (51) where a class 4 word precedes a monomoraic class 5 word. Association of the underlyingly floating H-tone in (51-b) and (51-c) implies a -0.9 violation of  $MAX-AL^{low}$  since the underlying  $M_{0.9}$  needs to deassociate and either a -0.8 violation of \*FLT-• (51-b) or a -0.9 violation of  $MAX-\bullet^{low}$  (51-c). Both these strategies result in a worse harmony score than simply leaving the  $H_1$  floating (51-a). Deletion of the H-tone in (51-d) avoids the \*FLT-• violation but higher-weighted  $MAX-\bullet^{high}$  is violated by -1.1. A floating H/S-tone that cannot find a host at the phrase layer since it does not precede a  $L_{0.1}$  will hence remain floating without any surface effect.

(51) Floating tone can't dock and remains floating at③

(cf. Appendix (81))

| $/M(H)^4/M^5$<br>$\begin{array}{cc} M_{0.9}H_{1.1} & M_{0.9} \\   &   \\ \mu_{0.8} & \mu_1 \end{array}$                 | $\text{MAX-}\bullet^{\text{high}}$<br>$\infty$ | $\text{*FLT-}\bullet$<br>379 | $\text{MAX-AL}^{\text{low}}$<br>320 | $\text{MAX-}\bullet^{\text{low}}$<br>115 | $\mathcal{H}$ |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------|-------------------------------------|------------------------------------------|---------------|
| a. $\begin{array}{cc} M_{0.8} \textcircled{H_1} & M_{0.8} \\   &   \\ \mu_{0.7} & \mu_{0.9} \end{array}$<br>[M] [M]     |                                                | -1                           |                                     |                                          | -379.0        |
| b. $\begin{array}{cc} M_{0.8} & H_1 \textcircled{M_{0.8}} \\   & \diagdown \\ \mu_{0.7} & \mu_1 \end{array}$<br>[M] [H] |                                                | -0.8                         | -0.9                                |                                          | -591.2        |
| c. $\begin{array}{cc} M_{0.8} & H_1 \\   & \diagdown \\ \mu_{0.7} & \mu_1 \end{array}$<br>[M] [H]                       |                                                |                              | -0.9                                | -0.9                                     | -391.5        |
| d. $\begin{array}{cc} M_{0.8} & M_{0.8} \\   &   \\ \mu_{0.7} & \mu_{0.9} \end{array}$<br>[M] [M]                       | -1.1                                           |                              |                                     |                                          | $-\infty$     |

(52) Weighting relations

|      |         |                                                                                      |     |                                  |
|------|---------|--------------------------------------------------------------------------------------|-----|----------------------------------|
| i.   | $a > b$ | $0.9 \times \text{MAX-AL}^{\text{low}}$                                              | $>$ | $0.2 \times \text{*FLT-}\bullet$ |
| ii.  | $a > c$ | $0.9 \times \text{MAX-AL}^{\text{low}} + 0.9 \times \text{MAX-}\bullet^{\text{low}}$ | $>$ | $1 \times \text{*FLT-}\bullet$   |
| iii. | $a > d$ | $1.1 \times \text{MAX-}\bullet^{\text{high}}$                                        | $>$ | $1 \times \text{*FLT-}\bullet$   |

Similarly to the threshold we established in (44) for  $\text{*hL}$ ,  $\text{*FLT-}\bullet$  is not important enough to cause deletion of originally underlying tones, it can only cause deletion of originally epenthetic tones. The relevant weighting relations are summarized again in (53). The second weighting relation in (53-b) adds the specification that deletion of a following underlying  $H_{0.9}/S_{0.9}$  is also excluded. It is clear that this strategy is even worse than deletion of a  $L_{0.9}/M_{0.9}$  illustrated in (51) since  $\text{MAX-}\bullet^{\text{high}}$  and  $\text{MAX-AL}^{\text{high}}$  have a higher weight than  $\text{MAX-}\bullet^{\text{low}}$  and  $\text{MAX-AL}^{\text{low}}$ .

(53) Overwriting to allow realization of a floating H/S at ③?

- a. Overwriting of originally epenthetic  $L_{0.1}$ , cf. (48)-i  
 $1 \times \text{*FLT-}\bullet > 0.1 \times \text{MAX-}\bullet^{\text{low}} + 0.1 \times \text{MAX-AL}^{\text{low}}$
- b. No overwriting of originally underlying  $T_{0.9}$ , cf. (52)-ii  
 $0.9 \times \text{MAX-AL}^{\text{low}} + 0.9 \times \text{MAX-}\bullet^{\text{low}} > 1 \times \text{*FLT-}\bullet$   
 $0.9 \times \text{MAX-AL}^{\text{high}} + 0.9 \times \text{MAX-}\bullet^{\text{high}} > 1 \times \text{*FLT-}\bullet$

Now we will turn to contexts where a floating H/S precedes multiple X-TBU's and hence the contexts where the mysterious M appears. Recall the empirical generalization that the floating H/S is realized on the rightmost X-TBU of the following word whereas all preceding X-TBU's are realized with M. We start with a context of a two-word combination  $/M(H)/ /X.X/$  where a floating H is hence followed by two TBU's associated to a weak  $L_{0.1}$  (55). The preceding tableaux already established that 1) an otherwise floating  $H_1$  will cause overwriting of an  $L_{0.1}$  in order to be realized (cf. tableau (47))

and 2) an associated H/S-tone will also cause complete deassociation+deletion of any following  $L_{0.1}$  in order to avoid a violation of \*hL (cf. tableau (45)). Tableau (55) hence only considers candidates where  $L_{0.1}$  is deleted and  $H_1$  is associated (but cf. the appendix (82) for a full tableau). Our most reasonable expectation for such a context is probably candidate (55-a) where the originally floating H associates to both otherwise tone-less TBU's. After all, tableau (40) established that H/S-tones spread to fill otherwise tone-less TBU's with a tone. However, there is a crucial activity difference between both contexts: In (40), an original  $/H_1/$  that already associated at the stem layer spreads and only -0.8 violations of \*SPR- $\bullet^{\text{high}}$  hence emerge at the phrase layer. In (54), however, the relevant H-tone is originally a strong  $H_{1.2}$  that still has 1 activity left at the phrase layer. And -1 violation of \*SPR- $\bullet^{\text{high}}$  is in fact worse than a -0.7 SPEC violation – spreading is hence sub-optimal in (55). If the H hence can only associate to one of the TBU's, our next-best expectation might be L-epenthesis as in candidate (54-b) to provide the other TBU with a tone. After all, we established in (25) that L-epenthesis repairs tone-less TBU's at the stem layer. But again, a crucial activity difference emerging in the HLT system blocks this repair. The moras have by now decayed so much that epenthesis is too expensive. More concretely: A -0.9 violation of SPEC at the stem layer was sufficient reason enough to tolerate -0.1 violation of DEP-T (and \*T) but the -0.7 violation at the phrase layer (54) is not a good enough reason: There is hence no L-epenthesis anymore at the last optimization cycle. One of the two TBU's in (54) thus has to remain tone-less since all strategies to provide it with a tone have failed. The only remaining question is which TBU remains tone-less and which one hosts the floating tone. The constraint deciding this is taken to be the markedness constraint (54) that always favors the rightmost possible association for an otherwise floating H. It is reminiscent of \*hL in that it penalizes [high] tones followed by a tone decline – in this case to a tone-less TBU that will phonetically equal a mid tone. This constraint excludes (54-d) and (54-c) emerges as optimal.

- (54) a. \*hø W= $\infty$   
 Assign  $-(n \times x)$  violations for every TBU associated to a [high] tonal root node  $^h\bullet_x$  that is immediately followed by a sequence of n tone-less tonal root nodes  $\bullet$ .

| $/M(H)^4/X.X.M^5$<br><div><math>\begin{array}{ccccc} M_{0.9} &amp; H_{1.1} &amp; L_{0.1} &amp; &amp; M_{0.9} \\   &amp; &amp; / \quad \backslash &amp; &amp;   \\ \mu_1 &amp; &amp; \mu_{0.8} \quad \mu_{0.8} &amp; &amp; \mu_1 \end{array}</math></div>                                                                                             | $\ast h\emptyset$<br>$\infty$ | DEP-T<br>810 | $\ast FLT-\bullet$<br>379 | MAX-AL <sup>low</sup><br>320 | $\ast hL$<br>152 | MAX- $\bullet^{low}$<br>115 | SPEC<br>95 | $\ast SPR-\bullet^{high}$<br>75 | $\mathcal{H}$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|---------------------------|------------------------------|------------------|-----------------------------|------------|---------------------------------|---------------|
| a.<br><div><math>\begin{array}{ccccc} M_{0.8} &amp; &amp; H_1 &amp; &amp; M_{0.8} \\   &amp; &amp; / \quad \backslash &amp; &amp;   \\ \mu_{0.9} &amp; &amp; \mu_{0.7} \quad \mu_{0.7} &amp; &amp; \mu_1 \end{array}</math><br/>[M] [H.H.M]</div>                                                                                                    |                               |              |                           | -0.2                         |                  | -0.1                        |            | -1                              | -150.5        |
| b.<br><div><math>\begin{array}{ccccc} M_{0.8} &amp; &amp; L_{0.1} &amp; H_1 &amp; M_{0.8} \\   &amp; &amp;   &amp; / \quad \backslash &amp;   \\ \mu_{0.9} &amp; &amp; \mu_{0.7} &amp; \mu_{0.7} &amp; \mu_1 \end{array}</math><br/>[M] [L.H.M]</div>                                                                                                |                               | -0.1         |                           | -0.2                         |                  | -0.1                        |            |                                 | -156.5        |
|  c.<br><div><math>\begin{array}{ccccc} M_{0.8} &amp; &amp; H_1 &amp; &amp; M_{0.8} \\   &amp; &amp;   &amp; / \quad \backslash &amp;   \\ \mu_{0.9} &amp; &amp; \mu_{0.7} &amp; \mu_{0.7} &amp; \mu_1 \end{array}</math><br/>[M] [<math>\emptyset</math>.H.M]</div> |                               |              |                           | -0.2                         |                  | -0.1                        | -0.7       |                                 | -142.0        |
| d.<br><div><math>\begin{array}{ccccc} M_{0.8} &amp; &amp; H_1 &amp; &amp; M_{0.8} \\   &amp; &amp;   &amp; / \quad \backslash &amp;   \\ \mu_{0.9} &amp; &amp; \mu_{0.7} &amp; \mu_{0.7} &amp; \mu_1 \end{array}</math><br/>[M] [H.<math>\emptyset</math>.M]</div>                                                                                   | -1                            |              |                           | -0.2                         |                  | -0.1                        | -0.7       |                                 | $-\infty$     |

## (56) Weighting relations

|      |         |                                            |
|------|---------|--------------------------------------------|
| i.   | $c > a$ | $1 \times *SPR-•^{high} > 0.7 \times SPEC$ |
| ii.  | $c > b$ | $0.1 \times DEP-T > 0.7 \times SPEC$       |
| iii. | $c > d$ | <i>harmonically bounded</i>                |

The association of floating H/S-tones to words with a multiply associated  $L_{0.1}$  hence results in tone-less TBU's. And those TBU's are taken to be realized with a phonetic default mid tone; the phonetic tone realization of the optimal structure in (57-d) is hence [M] [M.H] where the first M is the realization of an M-tone and the second M the realization of a tone-less TBU. This leaves us with only one portion of the floating tone generalization, namely the fact that a floating H/S-tone associates to the rightmost TBU of the following word and not the absolute rightmost tone-less TBU as we might now expect from everything shown in (55). That no word can be skipped by a floating tone is predicted from the phonologically predictable decay for moras that excludes the absolute final mora (cf. (23)). Originally word-final TBU's are thus always distinguishable from other TBU's at ② because they are slightly stronger. And this activity difference already predicts that any floating H/S will associate to them, simply because they would violate SPEC too much if left without any tonal specification. This is shown in tableau (57) where two bimoraic class 1 words follow a class 4 word. As in (55), all candidates the leave the H-tone floating or still realize any of the following  $L_{0.1}$  tones are omitted since we established that those have to be sub-optimal (but cf. the appendix (83) for a full tableau). From everything established in (55), we might expect (57-a) to become optimal that associates the H to the rightmost  $L_{0.1}$ -TBU and leaves all preceding TBU's unspecified, thus avoiding all \*hL and \*h $\emptyset$  violations. However, the subtle but crucial difference between sub-optimal (57-a) and optimal (55-c) is the fact that the latter only leaves TBU's with 0.7 activity tone-less whereas the former leaves

a stronger 0.9-TBU without a tone. And this gradient SPEC violation is in fact enough to exclude (57-a). Association of the H to only this stronger TBU in (57-b) fares better with respect to SPEC but is excluded by  $*h\emptyset$  since the H is now followed by tone-less TBU's. This violation is avoided in optimal (57-c) that spreads the H to all following TBU's. There is hence another crucial threshold arising from gradient violations: A -0.9 violation of SPEC is indeed enough to tolerate -1 violations of  $*SPR-\bullet^{high}$  whereas a -0.7 violation of SPEC is not enough to warrant a -1  $*SPR-\bullet^{high}$  violation. The latter was established in (55-a) and is also again shown in candidate (57-e) that associates the H to all following  $L_{0.1}$ -TBU's.

(57) Floating tone can skip TBU's but not words at ③

(cf. Appendix (83))

| /M (H)/ <sup>4</sup> /X.X/ <sup>3</sup> /X.X.M/ <sup>5</sup>                                                                                                                                                             |                                                                                                                                                                                                                                      | $*h\emptyset$ | DEP-T | $*FLT-\bullet$ | MAX-AL <sup>low</sup> | $*hL$ | MAX- $\bullet^{low}$ | SPEC | $*SPR-\bullet^{high}$ | $\mathcal{H}$ |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|----------------|-----------------------|-------|----------------------|------|-----------------------|---------------|
| $  \begin{array}{ccccccc}  M_{0.9} & H_{1.1} & L_{0.1} & & L_{0.1} & & M_{0.9} \\    & & / \quad \backslash & & / \quad \backslash & &   \\  \mu_1 & & \mu_{0.8} & \mu_1 & \mu_{0.8} & \mu_{0.8} & \mu_1  \end{array}  $ |                                                                                                                                                                                                                                      | $\infty$      | 810   | 379            | 320                   | 152   | 115                  | 95   | 75                    |               |
| a.                                                                                                                                                                                                                       | $  \begin{array}{ccccccc}  M_{0.8} & & & & H_1 & M_{0.8} \\    & & & &   &   \\  \mu_{0.9} & & \mu_{0.7} & \mu_{0.9} & \mu_{0.7} & \mu_{0.7} & \mu_1 \\  & & [M] & [\emptyset.\emptyset] & [\emptyset.H.M] & &   \end{array}  $      |               |       |                | -0.4                  |       | -0.2                 | -2.3 |                       | -369.5        |
| b.                                                                                                                                                                                                                       | $  \begin{array}{ccccccc}  M_{0.8} & & & & H_1 & & M_{0.8} \\    & & & &   & &   \\  \mu_{0.9} & & \mu_{0.7} & \mu_{0.9} & \mu_{0.7} & \mu_{0.7} & \mu_1 \\  & & [M] & [\emptyset.H] & [\emptyset.\emptyset.M] & &   \end{array}  $  | -1            |       |                | -0.4                  |       | -0.2                 | -2.1 |                       | $-\infty$     |
| c.                                                                                                                                                                                                                       | $  \begin{array}{ccccccc}  M_{0.8} & & & & H_1 & & M_{0.8} \\    & & & & / \quad \backslash & &   \\  \mu_{0.9} & & \mu_{0.7} & \mu_{0.9} & \mu_{0.7} & \mu_{0.7} & \mu_1 \\  & & [M] & [\emptyset.H] & [H.H.M] & &   \end{array}  $ |               |       |                | -0.4                  |       | -0.2                 | -0.7 | -2                    | -367.5        |
| d.                                                                                                                                                                                                                       | $  \begin{array}{ccccccc}  M_{0.8} & & & & H_1 & & M_{0.8} \\    & & & & / \quad \backslash & &   \\  \mu_{0.9} & & \mu_{0.7} & \mu_{0.9} & \mu_{0.7} & \mu_{0.7} & \mu_1 \\  & & [M] & [H.H] & [H.H.M] & &   \end{array}  $         |               |       |                | -0.4                  |       | -0.2                 |      | -3                    | -376          |

(58) Weighting relations

|      |         |                                                 |     |                                |
|------|---------|-------------------------------------------------|-----|--------------------------------|
| i.   | $d > b$ | $1.6 \times \text{SPEC}$                        | $>$ | $2 \times *SPR-\bullet^{high}$ |
| ii.  | $d > c$ | $1 \times *h\emptyset + 1.4 \times \text{SPEC}$ | $>$ | $2 \times *SPR-\bullet^{high}$ |
| iii. | $d > e$ | $1 \times *SPR-\bullet^{high}$                  | $>$ | $0.7 \times \text{SPEC}$       |

#### 4.4 Summary of the account: Gradient violations in HLT

The HLT account presented above relies on several gradient asymmetries where tones and TBU's with different activities behave differently in the phonology. These activity differences have three relevant sources: 1) predictable activity decay that predicts different behaviour at different layers (59-a+e), 2) lexical activity differences that predict morpheme-specific differences (59-b+f), and 3) minimal tone epenthesis predicts that epenthetic tones will always be phonologically distinguishable from non-epenthetic ones (59-c+d). The crucial asymmetries in phonological behaviour arising from these differences are summarized in (59) together with the central gradient weighting relations predicting them.

(59) Summary: Gradient constraint violations and phonological asymmetries

|    |                                                                                          |     |                                                                                                                              |                                                |
|----|------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| a. | L-epenthesis at ① but tone-less TBU's at ③.                                              |     |                                                                                                                              |                                                |
|    | $0.9 \times \text{SPEC}$                                                                 | $>$ | $0.1 \times \text{DEP-T} + 0.1 \times *T$                                                                                    | (25)                                           |
|    |                                                                                          |     | $0.1 \times \text{DEP-T} + 0.1 \times *T$                                                                                    | $>$ $0.7 \times \text{SPEC}$ (55)              |
| b. | Some H/S remain floating at ② and only associate at ③.                                   |     |                                                                                                                              |                                                |
|    | $1 \times *FLT-\bullet$                                                                  | $>$ | $0.1 \times \text{MAX-AL}^{\text{low}} + 0.1 \times \text{MAX-}\bullet^{\text{low}}$                                         | (29)                                           |
|    |                                                                                          |     | $1.1 \times *STRONG-\bullet + 0.1 \times \text{MAX-AL}^{\text{low}} + 0.1 \times \text{MAX-}\bullet^{\text{low}}$            | $>$ $1.1 \times *FLT-\bullet$ (36)             |
| c. | H/S-Spreading: *hL causes overwriting of epenthetic but not of underlying L's.           |     |                                                                                                                              |                                                |
|    | $0.8 \times *hL$                                                                         | $>$ | $0.1 \times \text{MAX-AL}^{\text{low}} + 0.1 \times \text{MAX-}\bullet^{\text{low}} + 0.7 \times *SPR-\bullet^{\text{high}}$ | (40)                                           |
|    |                                                                                          |     | $0.9 \times \text{MAX-AL}^{\text{low}} + 0.9 \times \text{MAX-}\bullet^{\text{low}} + 0.8 \times *SPR-\bullet^{\text{high}}$ | $>$ $1 \times *hL$ (42)                        |
| d. | Floating H/S overwrite epenthetic but not underlying L's.                                |     |                                                                                                                              |                                                |
|    | $1 \times *FLT-\bullet$                                                                  | $>$ | $0.1 \times \text{MAX-AL}^{\text{low}} + 0.1 \times \text{MAX-}\bullet^{\text{low}}$                                         | (47)                                           |
|    |                                                                                          |     | $0.9 \times \text{MAX-AL}^{\text{low}} + 0.9 \times \text{MAX-}\bullet^{\text{low}}$                                         | $>$ $0.8 \times *FLT-\bullet$ (51)             |
| e. | Floating H/S can skip word-internal TBU's but not word-final ones.                       |     |                                                                                                                              |                                                |
|    | $1 \times *SPR-\bullet^{\text{high}}$                                                    | $>$ | $0.7 \times \text{SPEC}$                                                                                                     | (55)/(57)                                      |
|    |                                                                                          |     | $0.9 \times \text{SPEC}$                                                                                                     | $>$ $1 \times *SPR-\bullet^{\text{high}}$ (57) |
| f. | 'Associated' H/S spread to avoid word-internal tone-less TBU's but 'floating' H/S don't. |     |                                                                                                                              |                                                |
|    | $0.7 \times \text{SPEC}$                                                                 | $>$ | $0.8 \times *SPR-\bullet^{\text{high}}$                                                                                      | (40)                                           |
|    |                                                                                          |     | $1 \times *SPR-\bullet^{\text{high}}$                                                                                        | $>$ $0.7 \times \text{SPEC}$ (55)              |

That the present account can be extended to more ZAC data is discussed in the appendix in section 6.1. And it is compared to possible alternative explanations in section 6.2 in the appendix.

## 5 Conclusion

The research on tonal morphology is starting to receive a renewed interest in the last decade (e.g. Lionnet et al., 2023) and is starting to also include more and more non-African languages into the discussion (e.g. Palancar and Léonard, 2016). An empirical goal of this paper is to emphasize the importance of the latter and to highlight the theoretical significance of the interesting tonal properties of a Zacatepec language which was extensively and carefully described by Villard (2015).

The analysis of ZAC presented in this paper makes two broad theoretical claims: For one, it argues for HLT and hence a more restrictive theory of the morpho-phonology interface. In contrast to classical stratal models, it is not based on co-grammars within a language but on the cyclic optimization within a single language. All interstratal differences within a language thus have to fall out from predictable activity decay: Phonological elements can loose activity at each optimization. In the ZAC account presented here, this mechanism predicts a variety of asymmetries: 1) L-epenthesis applies at the stem layer but not at the phrase layer, 2) some tones can already associate at the stem layer whereas others have to wait for the phrase layer optimization, and 3) floating tones can never skip a word.

That the complex tonology of ZAC falls out in HLT is particularly interesting because the model is more restrictive than any stratal account where no restrictions about possible co-grammars within



a language exist (Zimmermann, 2025; Tebay et al., 2025).

Secondly, this paper makes an important contribution to the phonetics-phonology interface since it explicitly relies on the co-existence of both a phonological and a different phonetic default within a language. Across the tonal literature, one can identify four different concepts of default tones that have been argued to exist in specific languages, summarized in (60). The novelty of the present proposal is now that multiple of those concepts can co-exist within the same language, namely concepts (60-A) and (60-C).

(60) A typology of default tones

|    | Assigned...                          | Properties                                                                                                            | Example                                                           |
|----|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| A. | in the lexical phonology             | phonologically active                                                                                                 | Pulleyblank (1986) on Dschang or Paster and Kim (2011) on Tiriki  |
| B. | in the post-lexical phonology        | phonologically inert for the lexical phonology but active for the post-lexical phonology                              | Hyman and Byarushengo (1984) on Haya or Pulleyblank (1986) on Tiv |
| C. | after/at the end of phonology        | phonologically inert and phonetically stable                                                                          | Pulleyblank (1986) on Yoruba or Mtenje (1987) on Chichewa         |
| D. | as a default phonetic interpretation | phonologically inert and phonetically unstable (=transitional function with either a linear or sagging interpolation) | Myers (1998) on Chichewa or McPherson (2011) on Tommo So          |

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## 6 Appendix

### 6.1 Extending the account to more data

This section discusses two possible extensions of the present model to more tonal data in Zacatepec Chatino: The putative floating tones 6.1.1 and tonal melodies that are not co-determined at the word layer 6.1.2.

#### 6.1.1 No floating L-tones

As was emphasized in subsection 4.2, the present account differs from the generalization in Villard (2015) and rejects the existence of floating L-tones assumed for classes 3 and 13. In contrast, I assumed that those classes are best analysed as /L<sub>1</sub>/ and /M<sub>1</sub>.S<sub>1</sub>.L<sub>1</sub>/ instead. In this subsection, I want to motivate this assumption further and give an outlook on how the unexpected effects of class 3 and 13 tones can be incorporated into the present account.

Villard (2015) emphasizes that floating L-tones behave drastically different from floating H and LS. The surface effect that class 3 and 13 words have on following words thus does not fall out from the association principles assumed for floating tones in the majority of contexts. All the surface changes triggered by a class 3 and 13 word are listed in (61) where the surface forms of mono-, bi- and trimoraic bases with all 15 tonal melodies are given after a word ending in a putative ‘floating L’. Recall from above that some words are unattested; those cells are marked in grey below.

(61) Changes triggered by the ‘floating L’

|    | (L) + μ              | (L) + μ.μ                | (L) + μ.μ.μ                  |
|----|----------------------|--------------------------|------------------------------|
| 1. | /X/<br>[ML]          | /XX/<br>[M.L]            | /XXX/<br>[M.M.L]             |
| 2. | /L (LS)/<br>[M (LS)] | /X.L (LS)/<br>[M.M (LS)] | /X.X.L (LS)/<br>[M.M.M (LS)] |
| 3. | /L (L)/<br>[M (L)]   | /X.L (L)/<br>[M.M (L)]   | /X.X.L (L)/<br>[M.M.M (L)]   |
| 4. | /M/<br>[M]           | /X.M/<br>[M.M]           | /X.X.M/<br>[M.M.M]           |
| 5. |                      | /X.M (H)/<br>[M.M (H)]   | /X.X.M (H)/<br>[M.M.M (H)]   |
| 6. |                      | /L.M/<br>[M.M]           | /X.L.M/<br>[M.M.M]           |
| 7. | /M (LS)/<br>[M (LS)] | /L.M (LS)/<br>[M.M (LS)] | /X.L.M (LS)/<br>[M.L.M (LS)] |
| 8. | /LH/<br>[LH]         | /X.LH/<br>[M.LH]         | /X.X.LH/<br>[M.M.LH]         |

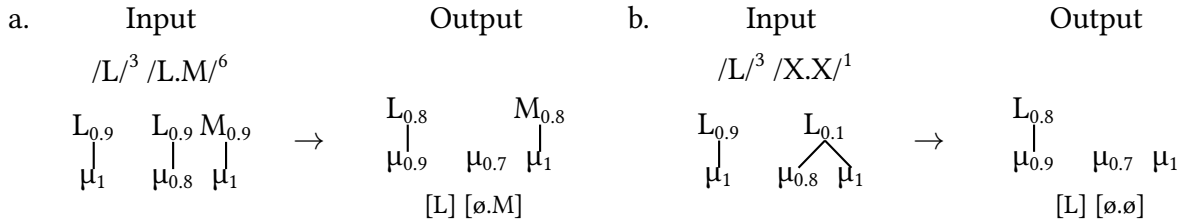
|     | (L) + μ      | (L) + μ.μ                | (L) + μ.μ.μ                  |
|-----|--------------|--------------------------|------------------------------|
| 9.  | /LS/<br>[LS] | /L.LS/<br>[L.LS]         | /X.L.LS/<br>[M.L.LS]         |
| 10. | /M/<br>[M]   | /M.M/<br>[M.M]           | /X.M.M/<br>[M.M.M]           |
| 11. | /H/<br>[H]   | /M.H/<br>[M.H]           | /X.M.H/<br>[M.M.H]           |
| 12. | /ML/<br>[ML] | /M.L/<br>[M.L]           | /M.M.L/<br>[M.M.L]           |
| 13. |              | /LS.L (L)/<br>[LS.L (L)] | /M.LS.L (L)/<br>[M.LS.L (L)] |
| 14. |              |                          | /M.H.M/<br>[M.H.M]           |
| 15. |              |                          | /L.M.L/<br>[L.M.L]           |

The changes in only 15 of the 45 contexts in (61) are consistent with the generalization that floating tones associate to the rightmost X-TBU of the following word and all potentially preceding X-TBU’s are realized with an M-tone. This is acknowledged in Villard (2015) where a set of five complex and class-specific linking rules is stated that account for the effects of the floating L-tone assumed there. In the majority of the 30 remaining contexts in (61), there are no additional L-tones

but rather only additional M-tones that surface either only on X-TBU's or replace underlying L-tones.

This appearance of so many additional M-tones is straightforwardly expected within the present proposal if we assume that the processes applying in (61) are not the result of associating a floating L but rather the deletion of L-tones on the following word. Since both class 3 and class 13 words end in an associated L-tone, a straightforward explanation is to assume that there are OCP-effects between adjacent L-tones. Given my assumption that M-tones are the absence of any tone, the appearance of so many additional M-tones in (61) is then simply the deletion of an L-tone and a resulting tone-less TBU. This is briefly shown for an underlying  $L_{0.8}$  in (62-a) and for a multiply associated epenthetic  $L_{0.1}$ -tone in (62-b) (=an 'X-TBU'). Both these L-tones are deleted since another L-tone precedes them. The TBU's have to remain tone-less since there is no way to provide them with another tone: Epenthesis is not only blocked at layer ③, it would only re-introduce the OCP-problem anyway. And there is naturally no H- or S-tone that could spread in this context: The TBU's are thus realized with a default mid interpretation in the phonetics.<sup>12</sup>

(62) OCP-triggered L-deletion at ③



This simple explanation is not yet sufficient, however since it mispredicts overapplication of L-deletion for all other tone melodies ending in L, namely classes 2, 12, 13, and 15. This issue of morpheme-specificity can straightforwardly be incorporated into the present model simply because gradient representations offer an inherent explanation for morpheme-specific effects: Different morphemes simply have differently active phonological elements. Since activity differences of the tonal root node were already essential to predict the tonal patterns discussed above, this morpheme-specific activity difference would affect register and melody features. More concretely, the melody and register features of the final L-tones in class 3 and 13 morpheme are taken to have an exceptionally strong activity whereas the register and melody features of other L-final morphemes have the default activity of 1.

The assumption of different activities for tonal features to encode the exceptional triggering property of class 3 and 13 can also be extended to explain the exceptional non-undergoer property of class 7, 9, and 15 that systematically resist L-deletion (cf. (61)). Those are assumed to have even stronger register and melody tonal features that make any deletion impossible since the MAX constraints penalizing feature deletion would be violated too much. This leaves only the behaviour of class 1 words in (61) unexplained. A possible representational solution is to assume that class 1 words are not tone-less after all but contain an L-tone which has a weakly active tonal root node. It is hence overwritten if a floating H or S precede and also if an associated H or S precede and shows the same behaviour as a tone-less TBU when it comes to floating tone association and tone spreading. A preceding L-tone with strongly active tonal features (=class 3 or 13), however, can't

<sup>12</sup>Note that the relevant OCP-constraint has to ban adjacent TBU's that are solely associated to L-tones; a class 3 or 13 word that precedes a word beginning with an LH or LS contour does not violate it (cf. (61-8+9)).

cause deletion because the class 1 L-tone has strongly active tonal features; absolutely parallel to the representation assumed for the non-undergoers in class 7, 9, and 15.

The assumption that all underlying phonological elements can potentially contrast their activity which is implicit in any HLT account hence offers an explanation for morpheme-specific exceptional behaviour which cross-cuts different processes. That a surface M-tone can be the phonetic default realization of a tone-less TBU in ZAC thus also offers a natural explanation for the many seemingly unexpected M-tones after class 3 and 13. They are taken to be the result of a morpheme-specific trigger for an OCP-repair that deletes both originally epenthetic and originally underlying L-tones.

### 6.1.2 Tonal melodies at the word layer

The account in section 4 is based on the assumption that all tonal morphemes are added at the word layer and the stem optimization hence only optimizes tone-less roots, an assumption that readily explains the tone patterns of verbs and inalienable nouns. There are, however, some words that are not obligatorily inflected: Alienable nouns, adverbs, some particles (e.g. a polar question marker). One might argue that it is the more economic and/or natural assumption that those already receive their tonal specification at the stem layer. The present account is perfectly compatible with this assumption. The same tone representations assumed in (27) hence predict exactly the same tonal behaviour at the phrase layer even if they are already added one layer earlier. This is sketched in (63) for a class 4 melody realized on a bimoraic word. All tones enter the phrase layer with 0.1 activity less than comparable tones that are only added at the word layer. For an underlying strong and thus floating tone, this means that it is in principle already weak enough to be associated at the word layer without any fatal violation of \*STRONG-•. However, there simply is no host for the floating tone at this point since it is only adjacent to a TBU that is already associated to an underlying /T<sub>0.9</sub>/ that a floating H/s cannot overwrite. It has hence to wait until the phrase layer to see whether it now precedes a weakly active originally epenthetic tone.

(63) Derivation: Tonal morpheme added at ①

| Input ①                         |   | Output ①/Input ②                                   |   | Output ②/Input ③                                 |   | Output ③                                           |
|---------------------------------|---|----------------------------------------------------|---|--------------------------------------------------|---|----------------------------------------------------|
| M <sub>1</sub> H <sub>1.2</sub> | → | L <sub>0.1</sub> M <sub>0.9</sub> H <sub>1.1</sub> | → | L <sub>0.1</sub> M <sub>0.8</sub> H <sub>1</sub> | → | L <sub>0.1</sub> M <sub>0.7</sub> H <sub>0.9</sub> |
| μ <sub>1</sub> μ <sub>1</sub>   |   | μ <sub>0.9</sub> μ <sub>1</sub>                    |   | μ <sub>0.8</sub> μ <sub>1</sub>                  |   | μ <sub>0.7</sub> μ <sub>1</sub>                    |

The grammar discussed above for ZAC indeed predicts the same tonal behaviour for structures as (63) and for ones that have their tonal melodies added only at the word layer. More concretely, all the weighting relations summarized in (59) also hold true if the constraints referring to tones are violated by 0.1 less/more. This is briefly shown in (64) that repeats the optimization from (57) but has now an initial class 4 noun that already had its tones added at the stem layer: They are hence already weaker than the comparable tones of the following words. Consequently, \*FLT-• is violated only by -0.9 if the H-topne remains unassociated (64-a), every L-toned TBU following the associated H only induces -0.9 violations of \*hL (64-b), and every additional association of the H-tone only induces -0.9 violations of \*SPR-•<sup>high</sup>. Nevertheless, the constraint weights correctly predict that the floating H<sub>0.9</sub> causes deletion of any following L<sub>0.1</sub>'s, associates to the next TBU with 0.9 activity, and spreads to any following ones. It hence behaves absolutely parallel to the stronger floating H-tone in (57) despite the slightly different constraint violations.

(64) Tableaux (57) with an initial weaker word

| $/M(H)^4/X.X^3/X.X.M^5$                                                                                                                | DEP-T | *FLT-• | MAX-AL <sup>low</sup> | *hL  | MAX-• <sup>low</sup> | *SPR-• <sup>low</sup> | SPEC | *SPR-• <sup>high</sup> | $\mathcal{H}$ |
|----------------------------------------------------------------------------------------------------------------------------------------|-------|--------|-----------------------|------|----------------------|-----------------------|------|------------------------|---------------|
| $M_{0.8}$ $H_1$ $L_{0.1}$ $L_{0.1}$ $M_{0.9}$<br>$\mu_1$ $\mu_{0.8}$ $\mu_1$ $\mu_{0.8}$ $\mu_1$                                       | 810   | 379    | 320                   | 152  | 115                  | 100                   | 95   | 75                     |               |
| a. $M_{0.7}$ $(H_{0.9})$ $L_{0.1}$ $L_{0.1}$ $M_{0.8}$<br>$\mu_{0.9}$ $\mu_{0.7}$ $\mu_{0.9}$ $\mu_{0.7}$ $\mu_1$<br>[M] [L.L] [L.L.M] |       | -0.9   |                       |      |                      | -0.2                  |      |                        | -361.1        |
| b. $M_{0.7}$ $H_{0.9}$ $L_{0.1}$ $L_{0.1}$ $M_{0.8}$<br>$\mu_{0.9}$ $\mu_{0.7}$ $\mu_{0.9}$ $\mu_{0.7}$ $\mu_1$<br>[M] [L.L] [L.L.M]   |       |        | -0.1                  | -2.7 |                      | -0.1                  |      |                        | -452.4        |
| c. $M_{0.7}$ $H_{0.9}$ $M_{0.8}$<br>$\mu_{0.9}$ $\mu_{0.7}$ $\mu_{0.9}$ $\mu_{0.7}$ $\mu_1$<br>[M] [H.H] [H.H.M]                       |       |        | -0.4                  |      | -0.2                 |                       |      | -2.7                   | -353.5        |
| e. $M_{0.7}$ $H_{0.9}$ $M_{0.8}$<br>$\mu_{0.9}$ $\mu_{0.7}$ $\mu_{0.9}$ $\mu_{0.7}$ $\mu_1$<br>[M] [∅.H] [H.H.M]                       |       |        | -0.4                  |      | -0.2                 |                       | -0.7 | -1.8                   | -352.5        |

Only one additional assumption is necessary. For one, if tonal melodies are already associated at the stem layer and thus before any epenthetic tones are inserted, an additional (low-weighted) gradient ALIGN-RIGHT constraint needs to ensure that underlying (and hence more active) tones are ideally clustered at the right edge whereas tone epenthesis for longer words applies at the left edge. This decides between, for example, a trimoraic [L.M.H] and \*[M.H.L] class 11 word or excludes [M.L(H)] for (63).

## 6.2 Discussion of alternative accounts

The description of the tonal system of ZAC in Villard (2015) provides phonological rules that derive the tonal surface forms and are immensely helpful in understanding the complex pattern. It is clear, however, that these rules are not intended as a formal account that derives the behaviour from general language-independent processes but to formalize the detailed empirical description. The HLT account presented in the last section aims to provide exactly such a general phonological formalization and presents a unified explanation for a number of interesting asymmetries in the tonal system of ZAC. I am not aware of an alternative account that similarly explains all the tonal behaviour from standard phonological mechanisms. Isolated aspects of the tonal system, however, might indeed be derivable from alternative explanations. In the following, I discuss alternatives to the positionally asymmetric weakening of moras in subsection 6.2.1, to the assumption that the mysterious M-tone is the phonetic realization of a tone-less TBU in 6.2.2, and to the assumption that H/S-spreading is a phonological process 6.2.3. It is clear that an alternative that rejects the HLT insights described above would need to rely on combining these (or similar) with additional completely orthogonal explanations to correctly predict the whole system.



### 6.2.1 Reference to prosodic words instead of mora decay

In the account above, the predictable decay for all non-final moras predicts the fact that floating tones cannot skip moras that were final at the word-layer. That moras which were final at the word layer are stronger than all other ones at the phrase layer hence translates a prosodic word boundary into a predictable activity difference. A seemingly obvious alternative is hence to simply refer to the prosodic word node instead. That the phonological derivation can refer to prosodic words is a seemingly uncontroversial assumption (Nespor and Vogel, 1986; Selkirk, 1986) and should be available both within and beyond a HLT system.

However, this specific restriction that floating tones cannot ‘skip’ a prosodic word is not trivial to implement since the floating tone is – well – floating and hence underlyingly not linearly ordered with respect to non-tonal elements. A constraint that correctly excludes the illegitimate floating to a prosodic word that is ‘too far away’ would hence need to refer both to prosodic words and to morphemic affiliation in order to track the underlying ‘birth place’ of the tone, exemplified in (65). It is clear that the formulation of (65) is complex at best in that it refers both to prosodic structure and to morpheme affiliation. And it remains an open question whether a constraint like (65) is a good candidate for a typologically motivated and hence language-independent constraint. The explanation in the HLT account avoids this pervasive problem of how to refer to the linear position of floating elements: It simply derives a special phonological representation for the ideal first landing site of the tone.

(65) **TONELOCALITY**

For every pair of tones T1 and T3 that have the same morphemic affiliation M:

Assign \* for every PrWd2 that intervenes between PrWd1 and PrWd3 if T1 is associated to a TBU within PrWd1 and T2 to a TBU within PrWd3 and no tone of the same morphological affiliation M is associated to a TBU that is part of PrWd2.

### 6.2.2 No tone-less TBU’s at the phrase layer

An obvious alternative explanation for the M-tone preceding the floating H- and S-tones is to assume that it is underlyingly present. This would make it unnecessary to rely on a phonetic default in addition to a phonological one and rely on the assumption that all TBU’s are fully specified after the phonological optimization. In the following, I discuss this alternative within the HLT model presented above. The underlying representations for tonal melodies 2, 4, and 7 would hence be as in (66-a) rather than (66-b). Absolutely parallel to the strong H and S, the strong M would remain unassociated at the word layer.

(66) **Alternative representations: Underlying M-tones**

|    | a. Alternative: Underlying M’s                                  | b. Present proposal                            |
|----|-----------------------------------------------------------------|------------------------------------------------|
| 2. | L <sub>1</sub> M <sub>1.2</sub> S <sub>1.2</sub>                | L <sub>1</sub> S <sub>1.2</sub>                |
| 4. | M <sub>1</sub> M <sub>1.2</sub> H <sub>1.2</sub>                | M <sub>1</sub> H <sub>1.2</sub>                |
| 7. | L <sub>1</sub> M <sub>1.2</sub> M <sub>1</sub> S <sub>1.2</sub> | L <sub>1</sub> M <sub>1</sub> S <sub>1.2</sub> |

This strong M-tone M<sub>1.2</sub> only has a chance to associate if a following word at the phrase layer starts with more than two TBU’s associated to a weak L<sub>0.1</sub> since association of the otherwise unassociated H or S takes precedence. This preference for association of a floating H/S over association of a floating

M/L already falls out from the grammar assumed here and is predicted from the respective weights of \*FLT-• and MAX-•<sup>low</sup> as was shown in tableau (33). That the M associates to multiple TBU's to avoid tone-less TBU's is also already correctly predicted, namely from the respective weight of SPEC and \*SPR-•<sup>low</sup>. The alternative underlying representation for a class 4 word following a trimoraic class 5 word is shown in (67) and it can be seen that the resulting output structure would be [M] [M.H.M], rather than [M] [ø.H.M], that was optimized in (55).

(67) Association of an originally strong  $M_{1,2}$

| Input ③                                                                                                                                                                       | Optimal output ③                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <hr/>                                                                                                                                                                         |                                                                                                                                                                         |
| /M (MH)/ <sup>4</sup> /X.X.M/ <sup>5</sup>                                                                                                                                    |                                                                                                                                                                         |
| $  \begin{array}{ccccc}  M_{0,9} & M_{1,1} & H_{1,1} & L_{0,1} & M_{0,9} \\    & & & / \quad \backslash &   \\  \mu_1 & & & \mu_{0,8} \quad \mu_{0,8} & \mu_1  \end{array}  $ | $  \begin{array}{ccccccc}  M_{0,8} & & M_1 & H_1 & M_{0,8} \\    & &   &   &   \\  \mu_{0,9} & & \mu_{0,7} & \mu_{0,7} & \mu_1 \\  [M] & & [M.H.M] & &   \end{array}  $ |

However, there is a serious complication with the assumption that the M-tone is part of the underlying representation. Recall from the discussion around tableau (33) that the present account implements the assumption that M- and L-tones that don't find a host at the word layer are deleted rather than left floating. The reason that these 'left-over' floating M- and L-tones at the left edge of words are excluded is simple: If they remain floating and enter the phrase layer, they are predicted to overwrite the weak  $L_{0,1}$  of a preceding class 1 word. This is shown in (68-b) where an alternative optimization for tableau (33) predicts candidate c. to become optimal at the word layer (68-a). If a monomoraic class 10 and 11 word now precedes such a floating M-tone, additional M-overwriting is mispredicted at the phrase layer (68-b).

(68) Alternative representations: M-tones remain floating

| a. Input ②                                                                                                                                        | Optimal output ②                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <hr/>                                                                                                                                             |                                                                                                                                                         |
| $  \begin{array}{ccc}  L_{0,1} & M_1 & H_1 \\    & & \\  \mu_1 & &   \end{array}  $                                                               | $  \begin{array}{ccc}  \textcircled{M_{0,9}} & H_{0,9} \\    &   \\  \mu_1 & \\  [H] &   \end{array}  $                                                 |
| b. Input ③                                                                                                                                        | Optimal output ③                                                                                                                                        |
| <hr/>                                                                                                                                             |                                                                                                                                                         |
| /X.X/ <sup>1</sup> /(M) H/ <sup>11</sup>                                                                                                          |                                                                                                                                                         |
| $  \begin{array}{ccc}  L_{0,1} & \textcircled{M_{0,9}} & H_{0,9} \\  / \quad \backslash & &   \\  \mu_{0,8} \quad \mu_1 & & \mu_1  \end{array}  $ | $  \begin{array}{ccccc}  L_{0,1} & M_{0,8} & & & H_{0,8} \\    &   & & &   \\  \mu_{0,7} & \mu_{0,9} & & & \mu_1 \\  [M.M] & [H] & & &   \end{array}  $ |

There is hence the conundrum that the strong floating M-tones of the alternative representations (66) need to remain floating at the word layer but the left-over ones in contexts as (33)/(68-b) need to be deleted. Within the present account, this prediction would be based on the weighting relations in (69) which are unsolvable.

- (69) a. No floating  $M_{0.9}$  at ②  
 $0.9 \times *FLT-\bullet + 0.9 \times *T > 1 \times MAX-\bullet^{low}$   
 b. Floating  $M_{1.1}$  at ②  
 $1.2 \times MAX-\bullet^{low} > 1.1 \times *FLT-\bullet + 1.1 \times *T$

The intuition that the mysterious M-tones are underlying is hence inherently incompatible with the account presented here. To successfully argue for this explanation, one hence has to find new accounts for all the other tonal patterns as well and show that they are compatible with the assumption of floating M-tones at the word layer. Crucially, the easy OCP-based explanation for the unexpected behaviour of class 3 and 13 (cf. 6.1.1) also vanishes as soon as one abandons the assumption that tone-less TBU's are phonetically realized as mid tones. And even if such an alternative is possible, the present alternative is representationally less complex since it does not need to rely on a mandatory M-tone preceding every floating tone.

### 6.2.3 Phonetic spreading

The account presented in this paper differs from the rule-based description given in Villard (2015) in many ways, primarily because my main goal is to derive the tonal surface forms from general and independently motivated principles and hence put the ZAC tonology into a larger theoretical context. The mysterious M-tone, for example, falls out from the timing of a phonological epenthesis process rather than from a language-specific realization rule for floating tones. This implies that two processes that are considered to be phonetic in Villard (2015) are taken to be phonological processes instead, namely H/S-spreading and L-epenthesis. The different divisions of work between phonology and phonetics are summarized in (70).

- (70) Processes in phonetics and phonology

|           | Villard (2015)                                                        | Current proposal                                                                        |
|-----------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Phonology | *Floating tone association<br>(6 language-specific association rules) | *Floating tone association<br>*L-Epenthesis<br>*H/S-Spread<br>*OCP-triggered L-deletion |
| Phonetics | * $\emptyset$ =L ("L-epenthesis")<br>*H/S-Spread                      | * $\emptyset$ =M                                                                        |

Interestingly, the arguments for treating L-epenthesis and H/S-spread as phonetic in Villard (2015) are rather based on their behaviour with respect to phonological principles rather than their phonetic properties. One argument is the different behaviour of spreading and floating tone association: "The environments for Spreading (and blocking of Spreading) and for floating tone realization are phonological, they both obey phonological rules, but their entity and their respective domain are different. As a result, Spreading is analyzed as a phonetic process (as in Pierrehumbert and Beckman (1988)), whereas floating tone behavior is considered phonological" (p.163). Although this observation is of course compatible with the assumption that one process is phonological and the other phonetics, it is by no means a necessary argument for this view since different phonological processes can naturally react differently to different phonological contexts.

Phonetically, spreading is described as "the persistence of a pitch target from preceding word

into subsequent tonally unspecified moras until the end of a utterance" (p.163) which creates "a slight contour (a slight downdrift of the F0)" (p.40). Crucially, this "slight downdrift" is not only observed for adjacent H- or S-toned TBU's that underwent spreading, it can in fact be observed for all sequences of TBU's with identical tone-specifications. It is hence similar to the situation often described as "declination" rather than "downdrift" (cf. Connell (2001) for a concise summary of the different definitions and uses of "downdrift" and "declination"). On p.201, for example, the pitch tracks for a trisyllabic class 1 word in isolation are given; hence a /X.X.X/ word whose TBU's are realized with a default L. The pitch contour shows a slight decline over the three TBU's. Crucially, however, such a decline can also be found for a class 3 word /X.X.L/ that ends in an underlying L-tone. The phonetic realization hence shows a predictable decline between TBU's with like tones but seemingly does not distinguish between underlying and epenthetic L-tones. That the latter are phonetic defaults is hence as compatible with the acoustic facts as the assumption that they are inserted phonologically.

There is thus no striking argument for either a phonetic or phonological characterization if one looks only at H/S-spread and epenthesis in isolation. The bigger picture of tonal patterns in ZAC, however, indeed provides an argument for treating them as phonological processes, namely the fact that the mysterious M-tones fall out as phonetic realizations of tone-less TBU's.

If H/S-spread, however, is taken to be a phonetic process, L-epenthesis also has to be a phonetic process; otherwise the spreading process would be expected to treat underlying and phonological default-L-tones identical and hence either over- or underapply. But if L-epenthesis is a phonetic process, the distinction into two fates for tone-less TBU's the present proposal rests on vanishes. Consequently, Villard (2015) assumes rather complex and language-specific realization rules for floating tones (cf. 2.3) to explain the appearance of M-tones before floating tones.

6.3 Full tableaux

(71) L-Epenthesis at ①

| $\mu_1$ $\mu_1$                                                                                        | $\infty$ MAX- $\bullet^{\text{high}}$ | $\infty$ MAX-AL $^{\text{high}}$ | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | 810 DEP-T | 380 *FLT- $\bullet$ | 320 MAX-AL $^{\text{low}}$ | 140 *hL | 115 SPEC | 110 MAX- $\bullet^{\text{low}}$ | 100 *SPR- $\bullet^{\text{low}}$ | 85 *SPR- $\bullet^{\text{high}}$ | 8 *T | $\mathcal{H}$ |
|--------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------|-----------------------------|-------------------------|-----------|---------------------|----------------------------|---------|----------|---------------------------------|----------------------------------|----------------------------------|------|---------------|
| a. $\mu_{0.9}$ $\mu_1$<br>[ $\emptyset.\emptyset$ ]                                                    |                                       |                                  |                             |                         |           |                     |                            |         | -1.9     |                                 |                                  |                                  |      | -218.5        |
| b. $\begin{array}{c} L_{0.1} \\ \swarrow \quad \searrow \\ \mu_{0.9} \quad \mu_1 \\ [L.L] \end{array}$ |                                       |                                  |                             |                         | -0.1      |                     |                            |         |          |                                 | -0.1                             |                                  | -0.1 | -91.8         |
| c. $\begin{array}{c} L_1 \\ \swarrow \quad \searrow \\ \mu_{0.9} \quad \mu_1 \\ [L.L] \end{array}$     |                                       |                                  |                             |                         | -1        |                     |                            |         |          |                                 | -1                               |                                  | -1   | -718.0        |
| d. $\begin{array}{cc} L_{0.1} & L_{0.1} \\   &   \\ \mu_{0.9} & \mu_1 \\ [L.L] \end{array}$            |                                       |                                  |                             |                         | -0.2      |                     |                            |         |          |                                 |                                  |                                  | -0.2 | -163.6        |

(72) Tonal melody association at ② : More TBU's than tones (class 11)

| $  \begin{array}{c}  L_{0.1} \quad M_1 \quad H_1 \\  \swarrow \quad \downarrow \quad \searrow \\  \mu_{0.9} \quad \mu_{0.9} \quad \mu_1  \end{array}  $                            | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T | 380 *FLT- $\bullet$ | 320 MAX-AL <sup>low</sup> | 140 *hL | 115 SPEC | 110 MAX- $\bullet$ <sup>low</sup> | 100 *SPR- $\bullet$ <sup>low</sup> | 85 *SPR- $\bullet$ <sup>high</sup> | 8 *T | $\mathcal{H}$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|-------|---------------------|---------------------------|---------|----------|-----------------------------------|------------------------------------|------------------------------------|------|---------------|
| a. $  \begin{array}{c}  L_{0.1} \quad (M_{0.9}) \quad (H_{0.9}) \\  \swarrow \quad \downarrow \quad \searrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  [L.L.L]  \end{array}  $ |                                         |                                 |                             |                         |       | -1.8                |                           |         |          |                                   | -0.2                               |                                    | -1.9 | -719.2        |
| b. $  \begin{array}{c}  L_{0.1} \\  \swarrow \quad \downarrow \quad \searrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  [L.L.L]  \end{array}  $                                 | -1                                      |                                 |                             |                         |       |                     |                           |         |          | -1                                | -0.2                               |                                    | -0.1 | $-\infty$     |
| c. $  \begin{array}{c}  L_{0.1} \quad H_{0.9} \\  \swarrow \quad \downarrow \quad \searrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  [L.L.M]  \end{array}  $                   |                                         |                                 |                             |                         |       |                     | -0.1                      |         |          | -1                                | -0.1                               |                                    | -1   | -160.0        |
| d. $  \begin{array}{c}  L_{0.1} \quad M_{0.9} \quad H_{0.9} \\  \swarrow \quad \downarrow \quad \searrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  [L.M.H]  \end{array}  $     |                                         |                                 |                             |                         |       |                     | -0.2                      |         |          |                                   |                                    |                                    | -1.9 | -79.2         |

(73) Tone association at ②: Same number of floating tones and TBU's (class 11)

|    | $\begin{array}{c} L_{0.1} \\ \swarrow \searrow \\ \mu_{0.9} \quad \mu_1 \end{array} \quad M_1 \quad H_1$                                                     | $\infty$ MAX- $\bullet^{\text{high}}$ | $\infty$ MAX-AL $^{\text{high}}$ | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL $^{\text{low}}$<br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet^{\text{low}}$<br>110 | *SPR- $\bullet^{\text{low}}$<br>100 | *SPR- $\bullet^{\text{high}}$<br>85 | *T<br>8 | $\mathcal{H}$ |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------|-----------------------------|-------------------------|--------------|------------------------|-------------------------------|------------|-------------|------------------------------------|-------------------------------------|-------------------------------------|---------|---------------|
| a. | $\begin{array}{c} L_{0.1} \\ \swarrow \searrow \\ \mu_{0.8} \quad \mu_1 \\ \text{[L.L]} \end{array} \quad \textcircled{M_{0.9}} \quad \textcircled{H_{0.9}}$ |                                       |                                  |                             |                         |              | -1.8                   |                               |            |             |                                    | -0.1                                |                                     | -1.9    | -709.2        |
| b. | $\begin{array}{c} L_{0.1} \\ \swarrow \searrow \\ \mu_{0.8} \quad \mu_1 \\ \text{[L.L]} \end{array}$                                                         | -1                                    |                                  |                             |                         |              |                        |                               |            |             | -1                                 | -0.1                                |                                     | -0.1    | $-\infty$     |
| c. | $\textcircled{L_{0.1}} \quad \begin{array}{c} M_{0.9} \quad H_{0.9} \\   \quad   \\ \mu_{0.8} \quad \mu_1 \\ \text{[M.H]} \end{array}$                       |                                       |                                  |                             |                         |              | -0.1                   | -0.2                          |            |             |                                    |                                     |                                     | -1.9    | -117.2        |
| d. | $\begin{array}{c} M_{0.9} \quad H_{0.9} \\   \quad   \\ \mu_{0.8} \quad \mu_1 \\ \text{[M.H]} \end{array}$                                                   |                                       |                                  |                             |                         |              |                        | -0.2                          |            |             | -0.1                               |                                     |                                     | -1.8    | -89.4         |
| e. | $\begin{array}{c} L_{0.1} \quad M_{0.9} \\   \quad   \\ \mu_{0.8} \quad \mu_1 \\ \text{[L.M]} \end{array} \quad \textcircled{H_{0.9}}$                       |                                       |                                  |                             |                         |              | -0.9                   | -0.1                          |            |             |                                    |                                     |                                     | -1.9    | -389.2        |

This tableau established that it is always a worse option to leave a deassociated  $L_{0.1}$  floating rather than deleting it altogether: In all following tableaux, I'll hence omit floating  $L_{0.1}$ -candidates.

(74) Tone association at ②: Fewer TBU's than tones (class 11)

| $L_{0.1} \quad M_1 \quad H_1$<br>$\mu_1$                                               | $\infty$ MAX- $\bullet^{\text{high}}$ | $\infty$ MAX-AL $^{\text{high}}$ | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T | *FLT- $\bullet$<br>380 | MAX-AL $^{\text{low}}$<br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet^{\text{low}}$<br>110 | *SPR- $\bullet^{\text{low}}$<br>100 | *SPR- $\bullet^{\text{high}}$<br>85 | *T<br>8 | $\mathcal{H}$ |
|----------------------------------------------------------------------------------------|---------------------------------------|----------------------------------|-----------------------------|-------------------------|-------|------------------------|-------------------------------|------------|-------------|------------------------------------|-------------------------------------|-------------------------------------|---------|---------------|
| a. $L_{0.1} \quad \textcircled{M_{0.9}} \quad \textcircled{H_{0.9}}$<br>$\mu_1$<br>[L] |                                       |                                  |                             |                         |       | -1.8                   |                               |            |             |                                    |                                     |                                     | -1.9    | -699.2        |
| b. $M_{0.9} \quad \textcircled{H_{0.9}}$<br>$\mu_1$<br>[M]                             |                                       |                                  |                             |                         |       | -0.9                   | -0.1                          |            |             | -0.1                               |                                     |                                     | -1.8    | -399.4        |
| c. $\textcircled{M_{0.9}} H_{0.9}$<br>$\mu_1$<br>[H]                                   |                                       |                                  |                             |                         |       | -0.9                   | -0.1                          |            |             | -0.1                               |                                     |                                     | -1.8    | -399.4        |
| d. $H_{0.9}$<br>$\mu_1$<br>[H]                                                         |                                       |                                  |                             |                         |       |                        | -0.1                          |            |             | -1.1                               |                                     |                                     | -0.9    | -160.2        |

This tableau established that it is always a worse option to leave a deassociated  $L_{0.8}/M_{0.8}$  floating rather than deleting it altogether: In all following tableaux, I'll hence omit floating  $L_{0.8}/M_{0.8}$ -candidates.



(75) Tone association at ②: Strong tones remain floating

| $  \begin{array}{c}  L_{0.1} \quad M_1 \quad H_{1.2} \\  \swarrow \quad \downarrow \quad \searrow \\  \mu_{0.9} \quad \mu_{0.9} \quad \mu_1  \end{array}  $                               | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL <sup>low</sup><br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet$ <sup>low</sup><br>110 | *SPR- $\bullet$ <sup>low</sup><br>100 | *SPR- $\bullet$ <sup>high</sup><br>85 | *T<br>8 | $\mathcal{H}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|--------------|------------------------|------------------------------|------------|-------------|--------------------------------------|---------------------------------------|---------------------------------------|---------|---------------|
| a. $  \begin{array}{c}  L_{0.1} \quad (M_{0.9}) \quad (H_{1.1}) \\  \swarrow \quad \downarrow \quad \searrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  \text{[L.L.L]}  \end{array}  $ |                                         |                                 |                             |                         |              | -2                     |                              |            |             |                                      | -0.2                                  |                                       | -2.1    | -796.8        |
| b. $  \begin{array}{c}  L_{0.1} \quad M_{0.9} \quad H_{1.1} \\  \downarrow \quad \downarrow \quad \downarrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  \text{[L.M.H]}  \end{array}  $ |                                         |                                 | -1.1                        |                         |              |                        | -0.2                         |            |             |                                      |                                       |                                       | -2.1    | $-\infty$     |
| c. $  \begin{array}{c}  L_{0.1} \quad M_{0.9} \quad (H_{1.1}) \\  \swarrow \quad \downarrow \quad \downarrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  \text{[L.L.M]}  \end{array}  $ |                                         |                                 |                             |                         |              | -1.1                   | -0.1                         |            |             |                                      | -0.1                                  |                                       | -2.1    | -476.8        |
| d. $  \begin{array}{c}  L_{0.1} \quad M_{0.9} \\  \swarrow \quad \downarrow \quad \downarrow \\  \mu_{0.8} \quad \mu_{0.8} \quad \mu_1 \\  \text{[L.L.M]}  \end{array}  $                 | -1.2                                    |                                 |                             |                         |              |                        | -0.1                         |            |             |                                      | -0.1                                  |                                       | -1      | $-\infty$     |

(76) H-Spreading at ③: Overwriting of  $L_{0.1}$

| $/H/^{11} /X.M/^5$<br>$\begin{array}{ccc} H_{0.9} & L_{0.1} & M_{0.8} \\   &   &   \\ \mu_1 & \mu_{0.7} & \mu_1 \end{array}$                 | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL <sup>low</sup><br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet$ <sup>low</sup><br>110 | *SPR- $\bullet$ <sup>low</sup><br>100 | *SPR- $\bullet$ <sup>high</sup><br>85 | *T<br>8 | $\mathcal{H}$ |
|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|--------------|------------------------|------------------------------|------------|-------------|--------------------------------------|---------------------------------------|---------------------------------------|---------|---------------|
| a. $\begin{array}{ccc} H_{0.8} & L_{0.1} & M_{0.8} \\   &   &   \\ \mu_{0.9} & \mu_{0.7} & \mu_1 \\ [H] & [L.M] & \end{array}$               |                                         |                                 |                             |                         |              |                        |                              | -0.8       |             |                                      |                                       |                                       | -1.7    | -125.6        |
| b. $\begin{array}{ccc} H_{0.8} & & M_{0.8} \\   & &   \\ \mu_{0.9} & \mu_{0.7} & \mu_1 \\ [H] & [\emptyset.M] & \end{array}$                 |                                         |                                 |                             |                         |              | -0.1                   |                              |            | -0.7        | -0.1                                 |                                       |                                       | -1.6    | -136.3        |
| c. $\begin{array}{ccc} H_{0.8} & & M_{0.8} \\   & \diagdown &   \\ \mu_{0.9} & \mu_{0.7} & \mu_1 \\ [H] & [H.M] & \end{array}$               |                                         |                                 |                             |                         |              |                        | -0.1                         |            |             | -0.1                                 |                                       | -0.8                                  | -1.6    | -123.8        |
| d. $\begin{array}{ccc} \textcircled{H_{0.8}} & L_{0.1} & M_{0.8} \\   &   &   \\ \mu_{0.9} & \mu_{0.7} & \mu_1 \\ [L] & [L.M] & \end{array}$ |                                         | -0.9                            |                             |                         |              | -0.8                   |                              |            |             |                                      |                                       |                                       | -1.7    | $-\infty$     |

(77) No H-Spreading at ③: No overwriting of  $L_{0.9}$

| $/H/^{11} \quad /L/^3$<br>$H_{0.9} \quad L_{0.9}$<br>$\mu_1 \quad \mu_1$                             | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL <sup>low</sup><br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet$ <sup>low</sup><br>110 | *SPR- $\bullet$ <sup>low</sup><br>100 | *SPR- $\bullet$ <sup>high</sup><br>85 | *T<br>8 | $\mathcal{H}$ |
|------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|--------------|------------------------|------------------------------|------------|-------------|--------------------------------------|---------------------------------------|---------------------------------------|---------|---------------|
| a. $\begin{array}{cc} H_{0.8} & L_{0.8} \\ \mu_{0.9} & \mu_1 \\ [H] & [L] \end{array}$               |                                         |                                 |                             |                         |              |                        |                              | -0.8       |             |                                      |                                       |                                       | -1.6    | -124.8        |
| b. $\begin{array}{cc} H_{0.8} & \\ \mu_{0.9} & \mu_1 \\ [H] & [H] \end{array}$                       |                                         |                                 |                             |                         |              |                        | -0.9                         |            |             | -0.9                                 |                                       | -0.8                                  | -0.8    | -461.4        |
| c. $\begin{array}{cc} H_{0.8} & \\ \mu_{0.9} & \mu_1 \\ [H] & [\emptyset] \end{array}$               |                                         |                                 |                             |                         |              |                        | -0.9                         |            | -1          | -0.9                                 |                                       |                                       | -0.8    | -508.4        |
| d. $\begin{array}{cc} \textcircled{H_{0.8}} & L_{0.8} \\ \mu_{0.9} & \mu_1 \\ [L] & [L] \end{array}$ |                                         | -0.9                            |                             |                         |              | -0.8                   |                              |            |             |                                      | -0.8                                  |                                       | -1.6    | $\infty$      |

## (78) H-Spreading at ③ across multiple words

| $/H/^{11} /X.X/^{1} /X.L/^{3}$<br> | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL <sup>low</sup><br>320 | *hl<br>140 | SPEC<br>115 | MAX- $\bullet$ <sup>low</sup><br>110 | *SPR- $\bullet$ <sup>low</sup><br>100 | *SPR- $\bullet$ <sup>high</sup><br>85 | *T<br>8 | $\mathcal{H}$ |
|------------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|--------------|------------------------|------------------------------|------------|-------------|--------------------------------------|---------------------------------------|---------------------------------------|---------|---------------|
| a.                                 |                                         |                                 |                             |                         |              |                        |                              | -3.2       |             |                                      |                                       | -0.1                                  | -1.8    | -472.4        |
| b.                                 |                                         |                                 |                             |                         |              |                        | -0.3                         | -0.8       |             | -0.2                                 |                                       | -2.4                                  | -1.6    | -446.8        |
| c.                                 |                                         | -0.9                            |                             |                         |              | -0.8                   |                              |            |             |                                      | -0.3                                  |                                       | -1.8    | $\infty$      |
| d.                                 |                                         |                                 |                             | -2.4                    |              |                        | -0.3                         |            | -2.3        | -0.2                                 |                                       |                                       | -1.6    | $-\infty$     |

(79) Floating tones overwrite  $L_{0.1}$  at ③

| $/M(H)^4/X.M.M/^{10}$<br>$\begin{array}{ccccc} M_{0.9} & H_{1.1} & L_{0.1} & M_{0.9} & M_{0.9} \\   & &   &   &   \\ \mu_1 & & \mu_{0.8} & \mu_{0.8} & \mu_1 \end{array}$        | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL <sup>low</sup><br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet$ <sup>low</sup><br>110 | *SPR- $\bullet$ <sup>low</sup><br>100 | *SPR- $\bullet$ <sup>high</sup><br>85 | *T<br>8 | $\mathcal{H}$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|--------------|------------------------|------------------------------|------------|-------------|--------------------------------------|---------------------------------------|---------------------------------------|---------|---------------|
| a. $\begin{array}{ccccc} M_{0.8} & \textcircled{H_1} & L_{0.1} & M_{0.8} & M_{0.8} \\   & &   &   &   \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} & \mu_1 \end{array}$<br>[M] [L.M.M] |                                         |                                 |                             |                         |              | -1                     |                              |            |             |                                      |                                       |                                       | -3.5    | -408.0        |
| b. $\begin{array}{ccccc} M_{0.8} & & L_{0.1} & M_{0.8} & M_{0.8} \\   & &   &   &   \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} & \mu_1 \end{array}$<br>[M] [L.M.M]                   | -1.1                                    |                                 |                             |                         |              |                        |                              |            |             |                                      |                                       |                                       | -2.5    | $-\infty$     |
| c. $\begin{array}{ccccc} M_{0.8} & H_1 & & M_{0.8} & M_{0.8} \\   & \diagdown & &   &   \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} & \mu_1 \end{array}$<br>[M] [H.M.M]               |                                         |                                 |                             |                         |              |                        | -0.1                         |            |             | -0.1                                 |                                       |                                       | -3.4    | -70.2         |

(80) Floating tones overwrite L<sub>0.1</sub> at ③, despite \*hL

| $/M(H)^4/X.L.M/^6$<br>$\begin{array}{ccccccc} M_{0.9} & H_{1.1} & L_{0.1} & L_{0.9} & M_{0.9} \\   & &   &   &   \\ \mu_1 & & \mu_{0.8} & \mu_{0.8} & \mu_1 \end{array}$             | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T | *FLT- $\bullet$ | MAX-AL <sup>low</sup> | *hL | SPEC | MAX- $\bullet$ <sup>low</sup> | *SPR- $\bullet$ <sup>low</sup> | *SPR- $\bullet$ <sup>high</sup> | *T   | $\mathcal{H}$ |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|-------|-----------------|-----------------------|-----|------|-------------------------------|--------------------------------|---------------------------------|------|---------------|
| a. $\begin{array}{ccccccc} M_{0.8} & \textcircled{H_1} & L_{0.1} & L_{0.8} & M_{0.8} \\   & &   &   &   \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} & \mu_1 \\ [M] & [L.L.M] \end{array}$ |                                         |                                 |                             |                         |       | -1              |                       |     |      |                               |                                |                                 | -3.5 | -408.0        |
| b. $\begin{array}{ccccccc} M_{0.8} & H_1 & & L_{0.8} & M_{0.8} \\   & \diagdown & &   &   \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} & \mu_1 \\ [M] & [H.L.M] \end{array}$               |                                         |                                 |                             |                         |       | -0.1            | -1                    |     |      | -0.1                          |                                |                                 | -3.4 | -194.2        |
| c. $\begin{array}{ccccccc} M_{0.8} & H_1 & & M_{0.8} \\   & \diagdown & &   \\ \mu_{0.9} & & \mu_{0.7} & \mu_{0.7} & \mu_1 \\ [M] & [H.H.M] \end{array}$                             |                                         |                                 |                             |                         |       | -1              |                       |     |      | -1                            |                                | -1                              | -2.6 | -535.8        |

(81) Floating tone can't dock and remains floating at ③

| $/M(H)^4 / M^5$<br>$M_{0.9}H_{1.1} \quad M_{0.9}$<br>$\mu_{0.8} \quad \mu_1$     | $\infty$ MAX- $\bullet^{\text{high}}$ | $\infty$ MAX-AL $^{\text{high}}$ | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL $^{\text{low}}$<br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet^{\text{low}}$<br>110 | *SPR- $\bullet^{\text{low}}$<br>100 | *SPR- $\bullet^{\text{high}}$<br>85 | *T<br>8 | $\mathcal{H}$ |
|----------------------------------------------------------------------------------|---------------------------------------|----------------------------------|-----------------------------|-------------------------|--------------|------------------------|-------------------------------|------------|-------------|------------------------------------|-------------------------------------|-------------------------------------|---------|---------------|
| a. $M_{0.8} \textcircled{H_1} M_{0.8}$<br>$\mu_{0.7} \quad \mu_{0.9}$<br>[M] [M] |                                       |                                  |                             |                         |              | -1                     |                               |            |             |                                    |                                     |                                     | -2.6    | 400.8         |
| b. $M_{0.8} H_1 \textcircled{M_{0.8}}$<br>$\mu_{0.7} \quad \mu_1$<br>[M] [H]     |                                       |                                  |                             |                         |              | -0.8                   | -0.9                          |            |             |                                    |                                     |                                     | -2.6    | -612.8        |
| c. $M_{0.8} H_1$<br>$\mu_{0.7} \quad \mu_1$<br>[M] [H]                           |                                       |                                  |                             |                         |              |                        | -0.9                          |            |             | -0.9                               |                                     |                                     | -1.8    | -401.4        |
| d. $M_{0.8} \quad M_{0.8}$<br>$\mu_{0.7} \quad \mu_{0.9}$<br>[M] [M]             | -1.1                                  |                                  |                             |                         |              |                        |                               |            |             |                                    |                                     |                                     | -1.6    | $-\infty$     |

(82) Floating tone association and tone-less TBU's at ③

| $/M(H)^4/X.X.M^5$<br><div><math>M_{0.9}</math> <math>H_{1.1}</math> <math>L_{0.1}</math> <math>M_{0.9}</math><br/><math>\mu_1</math> <math>\mu_{0.8}</math> <math>\mu_{0.8}</math> <math>\mu_1</math></div>                                                                                 | $\infty$<br>MAX- $\bullet$ <sup>high</sup> | $\infty$<br>MAX-AL <sup>high</sup> | $\infty$<br>*STRONG- $\bullet$ | $\infty$<br>*h $\emptyset$ | 810<br>DEP-T | 380<br>*FLT- $\bullet$ | 320<br>MAX-AL <sup>low</sup> | 140<br>*hL | 115<br>SPEC | 110<br>MAX- $\bullet$ <sup>low</sup> | 100<br>*SPR- $\bullet$ <sup>low</sup> | 85<br>*SPR- $\bullet$ <sup>high</sup> | 8<br>*T | $\mathcal{H}$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------|--------------------------------|----------------------------|--------------|------------------------|------------------------------|------------|-------------|--------------------------------------|---------------------------------------|---------------------------------------|---------|---------------|
| a. <div><math>M_{0.8}</math> <math>H_1</math> <math>M_{0.8}</math><br/><math>\mu_{0.9}</math> <math>\mu_{0.7}</math> <math>\mu_{0.7}</math> <math>\mu_1</math><br/>[M] [H.H.M]</div>                                                                                                        |                                            |                                    |                                |                            |              |                        | -0.2                         |            |             | -0.1                                 |                                       | -1                                    | -2.6    | -180.8        |
| b. <div><math>M_{0.8}</math> <math>L_{0.1}</math> <math>H_1</math> <math>M_{0.8}</math><br/><math>\mu_{0.9}</math> <math>\mu_{0.7}</math> <math>\mu_{0.7}</math> <math>\mu_1</math><br/>[M] [L.H.M]</div>                                                                                   |                                            |                                    |                                |                            | -0.1         |                        | -0.2                         |            |             | -0.1                                 |                                       |                                       | -2.7    | -177.6        |
|  c. <div><math>M_{0.8}</math> <math>H_1</math> <math>M_{0.8}</math><br/><math>\mu_{0.9}</math> <math>\mu_{0.7}</math> <math>\mu_{0.7}</math> <math>\mu_1</math><br/>[M] [<math>\emptyset</math>.H.M]</div> |                                            |                                    |                                |                            |              |                        | -0.2                         |            | -0.7        | -0.1                                 |                                       |                                       | -2.6    | 176.3         |
| d. <div><math>M_{0.8}</math> <math>H_1</math> <math>M_{0.8}</math><br/><math>\mu_{0.9}</math> <math>\mu_{0.7}</math> <math>\mu_{0.7}</math> <math>\mu_1</math><br/>[M] [H.<math>\emptyset</math>.M]</div>                                                                                   |                                            |                                    |                                | -1                         |              |                        | -0.2                         |            | -0.7        | -0.1                                 |                                       |                                       | -2.6    | $-\infty$     |
| e. <div><math>M_{0.8}</math> <math>\textcircled{H_1}</math> <math>L_{0.1}</math> <math>M_{0.8}</math><br/><math>\mu_{0.9}</math> <math>\mu_{0.7}</math> <math>\mu_{0.7}</math> <math>\mu_1</math><br/>[M] [L.L.M]</div>                                                                     |                                            |                                    |                                |                            |              | -1                     |                              |            |             |                                      | -0.1                                  |                                       | -2.7    | -411.6        |
| f. <div><math>M_{0.8}</math> <math>H_1</math> <math>L_{0.1}</math> <math>M_{0.8}</math><br/><math>\mu_{0.9}</math> <math>\mu_{0.7}</math> <math>\mu_{0.7}</math> <math>\mu_1</math><br/>[M] [H.L.M]</div>                                                                                   |                                            |                                    |                                |                            |              |                        | -0.1                         | -1         |             |                                      |                                       |                                       | -2.7    | -193.6        |
| g. <div><math>M_{0.8}</math> <math>L_{0.1}</math> <math>M_{0.8}</math><br/><math>\mu_{0.9}</math> <math>\mu_{0.7}</math> <math>\mu_{0.7}</math> <math>\mu_1</math><br/>[M] [L.L.M]</div>                                                                                                    | -1.1                                       |                                    |                                |                            |              |                        |                              |            |             |                                      | -0.1                                  |                                       | -1.7    | $-\infty$     |



(83) Floating H can not skip words but TBU's at ③

| $/M(H)^4/X.X/^3/X.X.M/^5$<br> | $\infty$ MAX- $\bullet$ <sup>high</sup> | $\infty$ MAX-AL <sup>high</sup> | $\infty$ *STRONG- $\bullet$ | $\infty$ *h $\emptyset$ | DEP-T<br>810 | *FLT- $\bullet$<br>380 | MAX-AL <sup>low</sup><br>320 | *hL<br>140 | SPEC<br>115 | MAX- $\bullet$ <sup>low</sup><br>110 | *SPR- $\bullet$ <sup>low</sup><br>100 | *SPR- $\bullet$ <sup>high</sup><br>85 | *T<br>8 | $\mathcal{H}$ |
|-------------------------------|-----------------------------------------|---------------------------------|-----------------------------|-------------------------|--------------|------------------------|------------------------------|------------|-------------|--------------------------------------|---------------------------------------|---------------------------------------|---------|---------------|
| a.                            |                                         |                                 |                             |                         |              |                        | -0.4                         |            | -2.3        | -0.2                                 |                                       |                                       | -2.6    | -435.3        |
| b.                            |                                         |                                 |                             | -2                      |              |                        | -0.4                         |            | -2.1        | -0.2                                 |                                       |                                       | -2.6    | $-\infty$     |
| c.                            |                                         |                                 |                             |                         |              |                        | -0.4                         |            | -0.7        | -0.2                                 |                                       | -2                                    | -2.6    | -421.3        |
| e.                            |                                         |                                 |                             |                         |              |                        | -0.4                         |            |             | -0.2                                 |                                       | -3                                    | -2.6    | -425.8        |
| f.                            |                                         |                                 |                             |                         |              | -1                     |                              |            |             |                                      | -0.2                                  |                                       | -2.8    | -422.4        |